

Highlights :

- The two basic sources of energy called exogenic (driven by Sun) and endogenic (driven by geothermal energy within the Earth) are explained.
- Classification of rocks based on the mode of formation into three basic types (igneous, sedimentary and metamorphic) are elaborated.

Introduction :

As a terrestrial planet, Earth's interior is divided into the core, the mantle and the crust. The crust and upper mantle are collectively known as the lithosphere, from where the tectonic plates originate and plays an important role in governing the rock cycle which results in the formation of rocks.

The Earth has two basic sources of energy. The energy that reaches the Earth from the Sun (solar energy) is exogenic meaning from an external source. Solar energy is converted into heat and absorbed by the atmosphere, hydrosphere and lithosphere. This heat drives the water cycle which further controls the exogenic processes. The other source of energy arising within the Earth as geothermal energy is called endogenic. Large amount of heat is trapped within the Earth. This energy is responsible for endogenic processes like tectonic activity.

It is in the lithosphere where the rocks are formed, recycled and re-formed. The nature of these rocks differ depending on the processes by which they are formed. The origin of any rock can be determined by its texture and composition which in turn forms the basis of rock identification and classification. Study of rocks in terms of its composition, texture, structure, origin, occurrence, alteration and association with other rocks is called as Petrology. The word petrology

is derived from the Greek word 'Petros' meaning rock and 'Logos' meaning science.

Rocks are naturally occurring aggregates of minerals. Based on their mode of formation, the rock forming processes are classified as :

- i) Igneous processes that involve melting, cooling and crystallisation of magma or lava.
- ii) Sedimentary processes involve weathering, erosion, deposition, and lithification of fragments of pre-existing rocks.
- iii) Metamorphic processes involve changes in texture, structure and / or mineral composition under different temperatures and / or pressure conditions generally in presence of chemically active fluids.

Over geological time, rocks are gradually transformed from one type to another and the process is described as 'Rock Cycle'.

The rock cycle is a model that describes the formation, breakdown, and reformation of a rock as a result of sedimentary, igneous, and metamorphic rock forming processes.

4.1 Exogenic and endogenic processes :

The dictionary meaning of the term exogenic is the process originating at or near the surface of the Earth, such as weathering and denudation. The sedimentary rocks, ore deposits and landforms are commonly formed by the exogenic processes. Endogenic processes originate within the Earth giving rise to the rocks, ore deposits and landforms that form due to tectonism, **diastrophism**, volcanism and **metamorphism**.

4.2 Extraterrestrial processes :

These include landforms formed by extraterrestrial activity such as impact of meteorites, e.g. the Lonar lake (Buldhana district, Maharashtra).

4.3 Genesis, classification and characteristics of rocks :

The early Earth was like a ball of fire with everything in molten state. While cooling, the outermost solid surface (crust) was first formed and this rock is called as an igneous ('Ignis' meaning fire) rock. Thus, all the rocks formed due to rapid or slow cooling of magma (within the subsurface) and lava (on the surface of the Earth) are called igneous rocks. A large variety of igneous rocks are therefore classified on the basis of rate of cooling of magma / lava and depth of formation.

In due course of time, the Earth experienced rainfall initiating the hydrological cycle (water cycle). This in turn initiated the phenomenon of weathering and denudation resulting in erosion and transportation of pre-existing rocks and their deposition in basins. Consolidation of the deposited material formed sedimentary rocks ('sedimentum' meaning settling). Several geological processes can result in rise in temperature and pressure causing the recrystallization of minerals of pre-existing rocks. These newly developed rocks are called as metamorphic (meta meaning 'After', morphic meaning 'Change in form') rocks.

The different approaches to the study of rocks are petrography, petrology, petrogenesis and lithology.

Petrology is a branch of geology that deals with description and systematic classification of rocks. The study of rocks as thin sections under microscope is called petrography.

Petrogenesis is a branch of petrology that deals with the origin and formation of rocks.

Lithology is a general terminology for description of rocks in hand specimen/or in the field, using characteristics like colour, texture and mineral composition.

4.3.1 Genesis, classification and characteristics of igneous rocks :

Igneous rocks are formed by cooling and consolidation of hot, molten rock material.

Magma :

Magma is a hot molten material formed at great depth. The temperature of the magma is around 800° to 1400° C. It is a hot melt of rocks or minerals. It is under great pressure at depth. Any factor which reduces the pressure will cause the eruption of the magma. Magma consists of volatile (gaseous) substances such as, water vapour, carbon dioxide, sulphur dioxide, etc. and non-volatile substances like oxides of silicon, aluminium, iron, calcium, sodium, potassium, magnesium, etc. When the same hot molten rock material erupts on the surface, it is known as lava.

Formation of igneous rocks :

When the magma present at great depth moves upward into the Earth's crust, both pressure and temperature decreases. Decrease in pressure within the crust will cause the intrusion of the magma onto the surface, as lava. Reduction of pressure allows the gases to escape into the surrounding rocks of the crust. The magma cools slowly and the whole mass of different minerals gives rise to igneous rocks. Types of igneous rocks thus formed, depends upon the original composition of the magma and its rate of cooling. Under slow rate of cooling, there is sufficient time and space for igneous rocks to form larger crystals (e.g. gabbro) whereas if the rate of cooling is rapid, fine grained rocks are formed. If the cooling is very fast, the rock melt does not develop minerals and forms a glassy rock, e.g. pitchstone, obsidian.

The igneous rocks, which are formed inside the Earth by injection of magma are known as intrusive or hypabyssal igneous rocks. These rocks are formed under a slow rate of cooling, are relatively medium grained. e.g. dolerite.

The rocks which are formed on the surface

by the eruption of volcanoes are known as extrusive or volcanic igneous rocks. These rocks are formed under conditions of rapid rate of cooling and are fine grained. e.g. basalt.

Classification of igneous rocks :

Igneous rocks are classified on the basis of genesis, chemical and mineralogical composition, petrographic characters and occurrences.

A simple classification is suggested based on chemical, mineral and textural characters. The chemical classification considers magmatic composition in which the percentage of felsic minerals (silica and aluminum bearing) and mafic minerals (iron and magnesium bearing) are present. Accordingly, the rocks having SiO_2 % > 66% are termed as acidic, those containing SiO_2 % : 65% - 55% are called intermediate, rocks containing SiO_2 % : 54% - 44% are basic while rocks with SiO_2 % < 44% are termed ultrabasic (Table 4.1).

Quartz is a mineral entirely comprising of SiO_2 , whereas SiO_2 can also occur in rocks as silicate compounds with other elements.

Classification based on mineral composition considers essential and accessory minerals, such as felsic and mafic rocks. Felsic rocks are light in color consisting of feldspars and quartz. Mafic rocks are dark colored, consisting of olivine, pyroxenes and amphiboles.

The general classification suggested for beginners is based on :

- 1) Mode of occurrence or depth of formation.
- 2) Percentage of quartz or silica
- 3) Presence of feldspars.

On this basis igneous rocks are classified as plutonic, hypabyssal and volcanic.

Plutonic igneous rocks :

Plutonic ('Pluto' meaning god of underworld) describes the igneous rock material whose origin is deep seated. Rate of cooling at this depth is slow. As a result, large and almost equal sized crystals are formed. Texture thus formed is called equigranular. The matrix of such rocks is completely composed of crystals which can be recognized by unaided eye. Such rocks are termed as holocrystalline ('holo' meaning complete) e.g. granite, syenite, gabbro.

Volcanic igneous rocks :

Magma generated within a magmatic chamber is suddenly and rapidly pushed upwards to the surface due to tectonic activity. As the magma moves upwards, fine grained minerals are formed due to rapid cooling. Before and during this movement, the earlier formed minerals grow larger in size. The admixture of fine and coarse grained minerals thus formed is described as porphyritic texture. Very rapid cooling does not permit crystallization, resulting into glassy (holohyaline) rock e.g. rhyolite, trachyte and basalt.

Table 4.1 : Tabular classification of igneous rocks

Silica %	Mode of occurrence or depth of formation		
	Plutonic	Hypabyssal	Volcanic
Acidic (> 66%)	Granite	Pegmatite	Rhyolite
Intermediate (65 - 55%)	Syenite		
Basic (54 - 44%)	Gabbro	Dolerite	Basalt
Ultrabasic (< 44%)	Dunite		

Hypabyssal igneous rocks :

The term hypabyssal is used for intrusive igneous rocks that crystallize at intermediate depths i.e., between plutonic and volcanic conditions. Some hypabyssal rocks contain coarse as well as fine grained minerals showing porphyritic relationship (e.g., pegmatite, dolerite).

Characteristics of igneous rocks :

- 1) These are first formed rocks by the cooling of magma or lava, hence are also called as primary rocks.
- 2) They are massive and unstratified.
- 3) They are non-fossiliferous.
- 4) They often occur as veins and fissures intruding into country rocks.
- 5) They exhibit baking or alteration effects along their contacts with country rocks.
- 6) They display an interlocking texture consisting of minerals, glass or both.
- 7) Lava flows may show gas cavities or amygdalae.
- 8) They are composed of pyrogenetic minerals formed at high temperature.
- 9) They exhibit equigranular / porphyritic / glassy texture.

4.3.2 Genesis, classification and characteristics of sedimentary rocks :

Sedimentary rocks are formed by weathering and erosion of the pre-existing rocks. The process of weathering and erosion is carried out by agents like sunlight, river, wind, glaciers, and oceans. The eroded products are transported and later deposited. These loose sediments on consolidation give rise to sedimentary rocks.

Genesis :

Sedimentary rocks are formed by **lithification** of weathered rock fragments that are physically transported and deposited. During transportation the fragments are abraded resulting in reduction in size. Such fragments are called clasts and the rock is referred to as **clastic** e.g., conglomerate and sandstone. Some of these

rocks are formed from chemical precipitation and crystallisation. Such rocks are termed as non-clastic e.g., limestone. Few sedimentary rocks are also formed by lithification of organic matter like shell fragments and are called as bioclastic e.g., fossiliferous limestone.

The process of deposition of loose fragments or clasts or rock debris is termed as sedimentation. It also includes material precipitated from solutions or water.

Classification :

Sedimentary rocks can be classified based on type of weathering, transportation and grain size (Table 4.2)

During transportation, water reacts with minerals and may form solutions. Under appropriate conditions precipitation takes place leading to formation of two types of deposits :

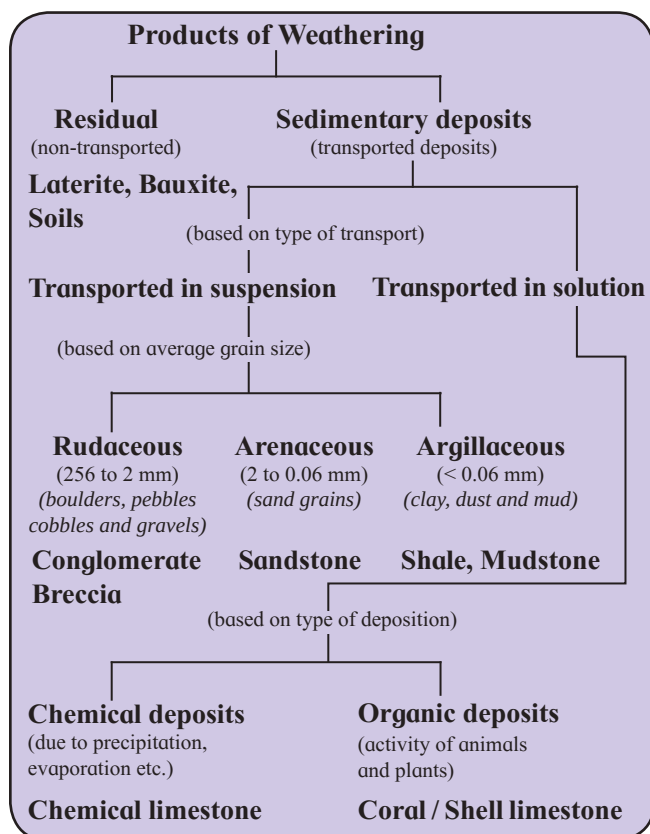
- 1) Chemical deposits : These deposits are formed by evaporation and precipitation e.g., limestone.
- 2) Organic deposits : Fragments of shells of animals and parts of plants are deposited and cemented to form fossiliferous and shell limestone.

During weathering, a rock may undergo *in-situ* chemical changes giving rise to residual deposits e.g., laterite, bauxite and soil.

Characteristics of sedimentary rocks :

- 1) They are derived from mechanical and / or chemical breakdown of pre-existing rocks.
- 2) Most of these rocks are layered or stratified.
- 3) These rocks show **clastic textures** with different cementing materials like silica, calcium carbonate, iron oxide, carbon or clay.
- 4) These rocks may contain fossils.
- 5) They may show primary structures like current bedding, ripple marks and mud cracks.
- 6) They may contain tracks, trails, burrows and imprints.

Table : 4.2 Classification of sedimentary rocks



4.3.3 Genesis, classification and characteristics of metamorphic rocks :

Rocks may undergo transformation due to change in temperature, pressure and presence of chemically active fluids.

Almost all pre-existing rocks are subjected to heat and pressure below the surface of the Earth. This brings about change in mineral composition and texture of original rock by recrystallisation. This results in the formation of new rock termed as metamorphic rock. Thus, the resulting rock shows change in mineral composition, texture and structure.

Genesis :

Metamorphism results from the change in physical and chemical environment of the original rock. These changes are because of different factors known as agents of metamorphism which are : temperature, pressure and chemically active fluids and gases.

1) Temperature : This is the most important agent of metamorphism. A rock begins

to change chemically at temperatures above 200°C. Temperature increases with increasing depth (due to geothermal gradient), intrusion of magma, plate tectonics and radioactive decay.

2) Pressure : Pressure increases with depth and due to tectonic movements. Rocks can be subjected to pressures of two kinds:

- Directional pressure
- Uniform or hydrostatic pressure.

Directional pressure is dominant within tectonically active zones. It is also termed as non-uniform pressure or stress. The effect of stress reduces the melting point of minerals and increases their solubility. As depth increases the directional pressure gradually changes into confining or uniform pressure.

3) Chemically active fluids and gases : These are derived from magma and are called as hydrothermal solutions. These fluids may contain carbon-dioxide, hydrofluoric acid, bromine, fluorine and water vapour in supersaturated state.

Classification of metamorphic rocks :

Metamorphic rocks are classified into three categories based on agents of metamorphism.

1) Cataclastic metamorphism : In this type the directional pressure plays an important role. There is no significant change in temperature. Since, directional pressure is the only operating agent, it mechanically breaks down the rock by crushing, fracturing and granulation. In this process no new mineral is formed e.g., slate.

2) Thermal metamorphism : In this kind of metamorphism there is increase in temperature, which is the dominant factor. The main source of heat is intrusion of magma. Along the contact between magma and host rock, recrystallization takes place (e.g., sandstone metamorphosed to quartzite and limestone metamorphosed to marble).

Table 4.3 : Genesis of metamorphic rocks

Original Rock	Types of Metamorphism (agent of metamorphism)	Product of Metamorphism	Structure
Argillaceous	Cataclastic (directed pressure or stress)	Slate	Slaty cleavage
Sandstone / Limestone	Thermal (temperature)	Quartzite/Marble	Granulose
Argillaceous (shale, mudstone) Basic igneous rock (basalt, gabbro)	Dynamothermal (directed pressure and temperature dominant)	Mica-garnet schist, Biotite schist, Biotite gneiss. Hornblende schist, Hornblende gneiss.	Schistose and Gneissose

3) Dynamothermal metamorphism : This metamorphism is also called regional metamorphism. Here heat and directional pressure are dominant. The rise in temperature and directional pressure is due to tectonic activity or magmatic intrusion. The temperature recrystallises the pre-existing minerals and directional pressure aligns the minerals perpendicular to direction of maximum pressure, thereby orienting the minerals to form schists and gneisses, e.g., hornblende gneiss, mica-garnet schist.

Characteristics of metamorphic rocks :

- 1) Metamorphic rocks are derived by the physical and chemical transformation of pre-existing (igneous, sedimentary or metamorphic) rocks.
- 2) They are formed in response to pronounced changes in the temperature, pressure and chemical environment.
- 3) Mineral transformation takes place in the solid state.
- 4) These rocks exhibit recrystallisation.
- 5) Metamorphic rocks show interlocking arrangement of minerals.
- 6) They exhibit granularity, schistosity and gneissosity.
- 7) Staurolite, kyanite, sillimanite, garnet are characteristic metamorphic minerals.
- 8) Metamorphic rocks are devoid of fossils

but are favoured to form economic ores and gemstones.

4.4 The rock cycle :

The rock cycle explains the evolution and transfer of igneous, sedimentary and metamorphic rocks. Rock components of the crust, slowly but consistently, are being changed due to different processes operating on the Earth.

The rock cycle is driven by two forces :

- 1) Earth's internal heat responsible for the endogenic processes.
- 2) The hydrological and other cycles on the Earth's surface powered by the sun (exogenic processes).

Due to internal forces generated within the Earth the intrusive rocks are uplifted and exposed to the surface. On their exposure to the atmosphere, these rocks get weathered, both physically and chemically. The weathered material after erosion gets transported by different agents and finally gets deposited as unconsolidated sediments along rivers, lakes and the ocean. With further addition of sediments, the weight (overburden) increases and the layers begin to subside. Eventually, with depth, the sediments are heated and partially melted to get mixed with the mantle material. This mixed mantle material may once again erupt as igneous rocks. Pre-existing rocks can be subjected to temperature and pressure and change to form metamorphic rocks. Thus, in nature, there appears to be a cyclic production, destruction and reproduction of all the three types of rocks. This appears to be a continuous process, ever since the

process of weathering and erosion started. This phenomenon is called the Rock cycle (Fig. 4.1)

Activity :

Collect various types of rocks from your area of residence and also from rock polishing shops. Identify their mineral contents and sort the rock samples into igneous, sedimentary and metamorphic rocks.

Summary :

The rocks are transformed from one type to another in response to the endogenic and exogenic processes by altering their mineralogy, texture and other physico-chemical properties.

Rocks are classified into three basic types and their intermediate forms are explained by the rock forming processes.

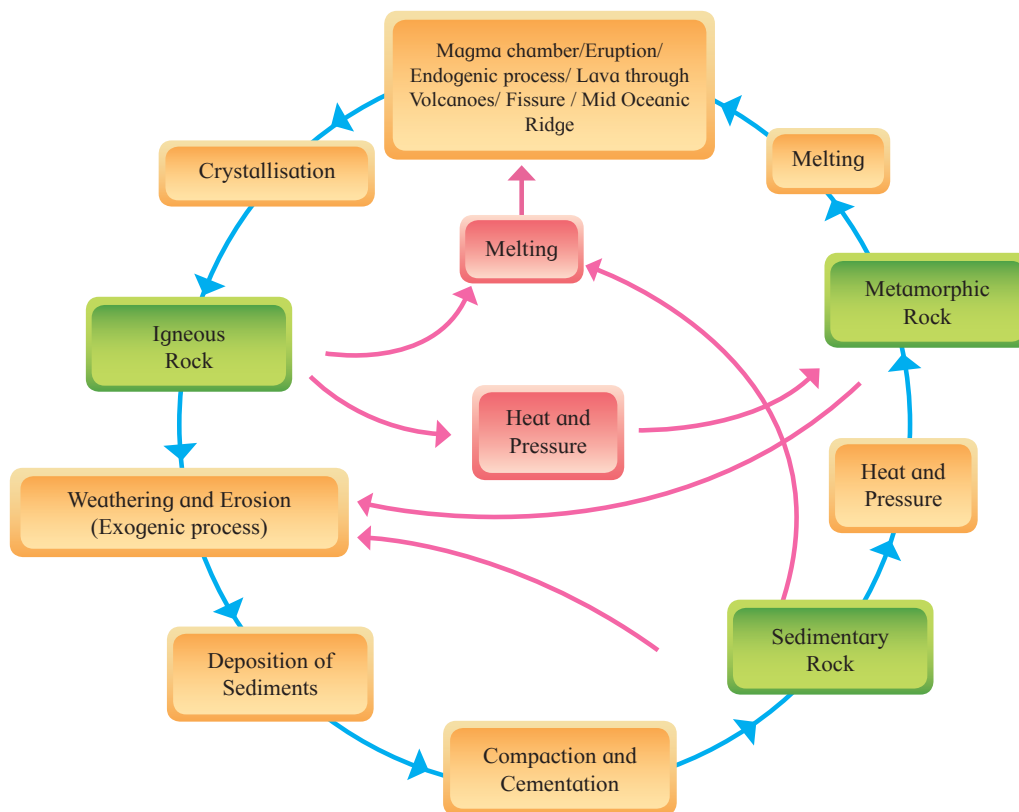


Fig. 4.1 : Rock cycle

EXERCISE

Q. 1. Fill in the blanks :

- 1) Metamorphic rocks characteristically show of material.
- 2) Cycle driven by exogenous and endogenous processes is known as

Q. 2. Multiple choice questions :

- 1) Sedimentary rocks are mainly formed by..... processes.
 - a) endogenous
 - b) exogenous

- c) extra terrestrial
- d) both exogenous and endogenous

- 2) If no crystallization takes place rock is developed.
 - a) holocrystalline
 - b) glassy
 - c) plutonic
 - d) Hypabyssal

Q. 3. Give geological terms :

- 1) Rocks that are classified on the basis of grain size are

- 2) The class of igneous rocks which exhibit equigranular texture are

Q. 4. Match the pairs :

1) Process	Product
A) Cataclastic	i) Granite
B) Thermal	ii) Slate
C) Dynamothermal	iii) Quartzite
D) Plutonic	iv) Mica schist

1) A-ii, B-iii, C-iv, D- i

2) A-iii, B-ii, C-i, D-iv

3) A-iv, B-i, C-iii, D-ii

4) A-ii, B-iv, C- iii, D- i

2) Types	Rocks
a) Rudaceous	i) Shell limestone
b) Organic deposits	ii) Conglomerate
c) Argillaceous	iii) Limestone
d) Chemical deposits	iv) Mudstone

1) a-iv, b-iii, c-i, d-ii

2) a-i, b-ii, c-iii, d-iv

3) a-iii, b-iv, c-ii, d-i

4) a-ii, b-i, c-iv, d-iii

Q. 5. Strike the odd one out :

- Plutonic, Hypabyssal, Volcanic, Organic deposits.
- Rudaceous, Arenaceous, Argillaceous, Hypabyssal
- Cataclastic, Thermal, Dynamothermal, Volcanic.

Q. 6. Eliminate the incorrect statement :

- The crystallisation of magma is not governed by the
 - Temperature and pressure
 - Composition of magma
 - Visco sity of magma
 - Geographic position of the magma on the Earth's surface
- Plutonic igneous rocks are
 - Deep seated and have moderate temperature pressure conditions.
 - Near the Earth's surface and possess high temperature pressure conditions.

- Deep seated with high temperature pressure and fast cooling conditions.
- Deep seated with high temperature pressure and slow cooling conditions.

Q. 7. Distinguish between :

- Igneous and Metamorphic rocks.
- Volcanic Igneous and Plutonic Igneous rocks.
- Rudaceous and Arenaceous Sedimentary rocks.
- Cataclastic and Thermal metamorphism.

Q. 8. State whether True or False and justify in either case :

- Hypabyssal rocks contain coarse and fine grained minerals.
- Sedimentary rocks are layered or stratified
- Metamorphic rocks exhibit recrystallization.

Q. 9. Write short notes on :

- Characteristics of igneous rocks.
- Classification of sedimentary rock.
- Genesis of sedimentary rocks.
- Genesis of metamorphic rocks.

Q. 10. Give geological reasons :

- Plutonic igneous rocks usually exhibit coarse grained texture.
- Volcanic rocks exhibit fine grained texture.
- No new minerals are formed in cataclastic metamorphism.
- Some minerals are formed in dynamothermal metamorphism.

Q. 11. Answer in detail :

- Explain the process of transformation of igneous and metamorphic rocks into sedimentary rocks.
- Explain the role of endogenic and exogenic processes in conversion of one rock type into another.
- Explain how metamorphic rocks can be converted to sedimentary rocks.



Highlights :

- Nomenclature and classification of minerals.
- Characteristic features of minerals.
- Physical properties for mineral identification.

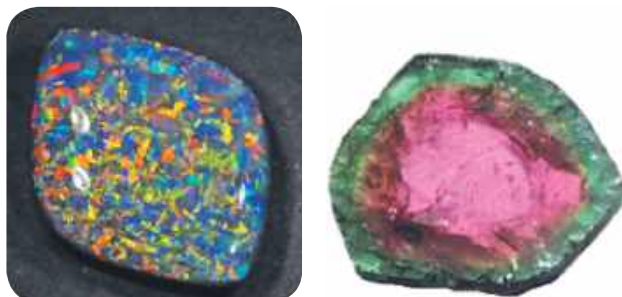
Introduction :

Rocks are fundamental units of the Earth's crust and study of Earth begins with the study of rocks. A close examination of rocks reveals that they are made up of smaller units known as minerals. Minerals may vary in colour, hardness, density, crystal form, crystal size, transparency, composition, occurrence and abundance.

Some minerals are radioactive (Uraninite) while others are magnetic (Magnetite) (Fig. 5.1).

**Fig. 5.1 : Uraninite****Magnetite**

Minerals such as fluorite glow with a vibrant luminescence after exposure to ultraviolet light. Feldspar and quartz are most abundant minerals found in the Earth's crust. Mankind has a desire for many minerals and gemstones due to their strikingly beautiful colors. Some specimens of opal exhibit rainbow-like colors while some specimens of the gemstone, tourmaline show progressions of watermelon like green to pink colours (Fig. 5.2).

**Fig. 5.2 : Opal****Tourmaline**

Few minerals are deceiving and many amateur prospectors have been tricked into thinking that they have found gold but have instead found pyrite (Iron disulphide) otherwise known as "Fools Gold" (Fig. 5.3 A) due to its uncanny resemblance to the precious metal.

**Fig. 5.3 : A) Pyrite (fool's gold)****B) Diamond**

Diamond, the hardest natural substance found on Earth, has long been priced as the most desirable of all minerals due to its spectacular interaction with light. Majority of diamonds are impure and not suitable as gemstones. They, however, can be used in the industry as cutting tools (Fig. 5.3 B).

5.1 Mineral nomenclature :

Name of any mineral is its identity. The names of many minerals are derived from antiquity. Most of the earlier mineral names were based on their physical properties like colour, density, etc. Later with the discovery of new elements, many new minerals were named. A set of guidelines formulated by the Commission on New Minerals and Mineral Names (CNMMN) under the International Mineralogical Association (IMA) are followed in naming minerals. The master list of minerals prepared by CNMMN in November 2018 has 5,389 minerals.

Many minerals are named after famous personalities or scientists who discovered or analysed them. Moreover, there are names of minerals that convey information about the region where they have been found. Some are quite specific about the location. Some mineral names convey useful information about the mineral itself and are based on the chemical

composition, colour, crystal form, hardness, lustre or other properties. Majority of mineral names end in 'ite' which is derived from the Greek word lithos (from its adjective form-ites) meaning rock or stone.

5.1.1 Minerals named after person :

The first mineral which was named after a person is Prehnite. It was named after Col. von Prehn in 1790. Since then, the naming of minerals after people became a common practice. Few other commonly occurring minerals which have been named after persons are Biotite (French mineralogist Jean-Baptiste Biot), Dolomite (French mineralogist, D.de Dolomieu), Heulandite (German mineral collector and dealer John Henry Heuland).

People who have two minerals named after them :

1) Marie Sklodowska Curie :

- a) Sklodowskite
- b) Curite (named after Marie and her French physicist husband Pierre).

2) Neil Alden Armstrong : American, astronaut, first human being to land on the moon's surface.

- a) Armstrongite
- b) Armalcolite (named after Neil Armstrong and other Apollo 11 crew members Aldrin and Collins).

Minerals named after persons of Indian origin :

1) Radhakrishnaite : Named after Bangalore Puttaiya Radhakrishnan (1918 - 2012) Indian Mineralogist, who was Director Mysore Geological Department.

A mineral found in Kolar Gold field, Karnataka.

2) Rustumite : Named after Rustum Roy (1924 - 2010) B.Sc and M.Sc in Chemistry from Patna University and Ph.D in ceramic science and Professor at Pennsylvania University, specialist in crystal chemistry of minerals and synthetic materials.

5.1.2 Minerals named after a place :

Common minerals named after the place from where they were first reported from or

are abundantly found are Amazonite (named after area near the Amazon River), Labradorite (named after Labrador province, Canada) and Muscovite (named after the Russian province Muscovy).

5.1.3 Minerals named after their chemical composition :

Common mineral names derived from their chemical composition are Calcite (derived from the Greek word 'Chalx' which means burnt lime), Cavansite (named after its elemental contents of Calcium, Vanadium and Silicon), Halite (Greek word 'hals' which means salt).

5.1.4 Minerals named after physical properties :

1) **Colour** : Some of the common minerals named after their color are Beryl (Greek word 'beryllos' which means a green stone), Hematite (after Greek word haimetites, meaning blood-like), Kyanite (Greek word 'kyanos' meaning blue).

2) **Lustre** : Stilbite-Named after German word 'stilbein' which means 'to shine' because of its brilliant lustre. It is a member of the zeolite group of minerals, commonly found as a secondary mineral in basalt.

3) **Cleavage** : Orthoclase-Named after the Greek words 'ortho' and 'klao' meaning 'straight' and 'cleave'. It breaks due to two cleavages meeting at right angles.

4) **Hardness** : Diamond-Named after the Greek word 'adamas' which means 'invincible' because of its hardness.

5) **Sp. gravity** : Barite-Named after the Greek word 'barys' which means 'heavy' because of its high specific gravity. It is also known as heavy spar.

6) **Magnetism** : Magnetite-Named for its strong magnetic property. It is an important ore of Iron.

5.1.5 Minerals named after Sanskrit word :

1) **Corundum** : Named after the Sanskrit word 'Kuruvind' which means 'ruby'.

2) **Opal** : Named after Sanskrit word 'Upala' which means 'gem'.

5.1.6 Mineral names from antiquity :

Many mineral names have been in existence from antiquity like Quartz, Galena, Gypsum etc.

5.2 History of Mineralogy :

The history of mineralogy goes back to Aristotle the Greek philosopher (384 - 322 BCE) who was one of the first to write about minerals and their properties in the western tradition.

Theophrastus, the Greek philosopher (372 - 287 BCE) is known for the first written work on minerals. He wrote a book on mineralogy called 'Concerning Stones'. Pliny the Elder, Roman naturalist (23 - 79 CE) recorded the mineralogical ideas of his time. He wrote an encyclopaedic work 'Natural History'. Agricola Georgius, a German scholar and scientist (1494-1555) was a town physician and considered as the 'Father of Mineralogy'. His book 'De re metallica' published in 1556, is about the mining practices and metallurgy techniques of his time.

5.3 Mineralogy - Salient points :

- 1) Elements are the building blocks of minerals while minerals are the building blocks of rocks.
- 2) There are more than 5000 mineral species approved by the International Mineralogical Association (IMA)
- 3) Silicate minerals compose over 95% of the Earth's crust. Silicon (27.7%) and Oxygen (46.6%) constitute approximately 75% of the Earth's crust.
- 4) Eight elements account for most of the key components of minerals due to their abundance in the crust. These eight elements constitute more than 98% of the crust by weight. These eight elements in decreasing order of abundance are Oxygen, Silicon, Aluminium, Iron, Magnesium, Calcium, Sodium and Potassium.
- 5) Differences in crystal structure and chemistry greatly influence the physical properties of minerals. The carbon allotropes diamond and graphite have vastly different properties.

5.4 Mineral characteristics :

Five criteria essential for classifying a substance as a mineral are as follows :

1) A mineral has to occur naturally :

Substances produced artificially by humans are not considered as minerals. With the help of modern technology, many compounds can be produced in the laboratory by artificially simulating the conditions under which they are formed in nature, e.g. precious and semi-precious stones such as topaz, corundum and even diamond can be produced synthetically. Cubic Zirconia also known as American diamond is a synthetic gem variety and does not qualify as a mineral.

2) A mineral has to be formed by inorganic processes :

Substances originating from plants (amber, rudraksha etc) or animal (ivory, pearls, corals etc) cannot be termed as minerals. Calcite or aragonite which is formed inorganically due to the processes of evaporation or crystallization are minerals, but the same substances deposited by the activity of organisms (mollusk shells) do not qualify as minerals.

3) A mineral has to be in a solid state :

Accordingly, water which is in liquid form is not a mineral while ice which is in solid form, is termed as a mineral. Therefore mercury, petroleum etc. also cannot be considered as minerals.

Do you know?

Natural oil (Petroleum) although occurs naturally, is liquid in nature and is formed by organic processes, is dealt with separately and not in mineralogy.

4) A mineral must have a definite chemical composition or a definite range of chemical composition :

Quartz crystals collected from any part of the Earth will always have the same chemical composition i.e SiO_2 . Many minerals may exhibit a range of chemical composition, for e.g. plagioclase feldspars form a wide

range of minerals from Na rich (albite) to Ca rich (anorthite) varieties.

- 5) A mineral must possess an ordered internal atomic structure :** Definite physical and chemical characters will be exhibited by a substance if its atoms are orderly arranged in a geometric pattern. Such substances qualify as minerals. For example natural glass which is formed by faster cooling of lava is not a mineral because it lacks an ordered internal atomic structure.

Activity :

Using the above significant characteristics of minerals, derive a holistic definition for a mineral.

5.5 Physical properties of minerals :

The physical properties vary from mineral to mineral depending on their chemical composition and crystal structure. Minerals possess widely different physical properties such as form, colour, streak, lustre, fracture, cleavage, hardness, specific gravity etc. The study of these physical properties is the first step in identification of minerals. Some properties like radioactivity, specific gravity, magnetism, electrical and thermal properties may need instruments for measurement.

Some of the original properties of the minerals may be partly or completely altered due to their exposure to air and water which causes corrosion or rusting. Hence, whenever geological studies are to be carried out, it is important to obtain fresh, *in-situ* sample.

5.5.1 Characters depending upon light :

1) Colour : Colour is the first property one notices in a mineral. It is also one of the reasons that attracts people to minerals, especially to gem stones. However, colour is not a very reliable property to be used in the identification of minerals. It usually confuses an inexperienced collector into making an incorrect identification.

It is important to know what causes colour in minerals in order to understand this mineral property.

Colour is determined by its wavelength. When pure white light (containing all wavelengths of visible light) enters a mineral, some of the wavelengths might be absorbed while other wavelengths are emitted. If this happens, then the light that is transmitted through the mineral will no longer be white but will have some colour. Colour in minerals is caused by the absorption or lack of absorption of various wavelengths of light.

Colourless minerals (transparent and translucent) like quartz, calcite, fluorite can exhibit certain colours due to the presence of transition elements. These transition elements cause a mineral to exhibit a certain colour if they are part of the chemistry of the mineral. The most common chromophoric (colour causing) transition elements with unfilled electron shells are Ti, V, Cr, Mn, Fe, Co, Ni, Cu. However, if there is just a trace of these elements, they still can strongly influence the colour of the mineral. Copper usually produces green and blue colours. Iron is known for the red and yellow colours that it typically produces. However, almost any element can be responsible for a characteristic colour. Thus, minerals of same composition can exhibit different colours e.g. quartz (Fig. 5.4), calcite (Fig. 5.5), fluorite (Fig. 5.6) etc. Minerals of different compositions can also exhibit the same colour e.g. calcite (Fig. 5.7 A), natrolite, (Fig. 5.7 B) and milky quartz (Fig. 5.7 C) all exhibit white colour. Hence, colour cannot be considered as the most diagnostic property for identifying minerals.

Quartz varies widely in colour, due to minor (parts per billion) impurities and even defects in its crystalline structure. Pure quartz has no colour. Varieties of quartz, shown in figure 5.4 include : amethyst (purple), smoky quartz (brown to black), citrine (yellow), rosy quartz (pink), rock crystal (clear).



Amethyst



Smoky quartz



Citrine



Rosy quartz



Rock crystal

Fig. 5.4 : Varieties of quartz showing different colours due to presence of chromophoric elements



Fig. 5.5 : Variety of colours exhibited by Calcite



Fig. 5.6 : Variety of colours exhibited by Fluorite (CaF_2)



A) Calcite (CaCO_3)



B) Natrolite ($\text{Na}_2\text{Al}_2\text{Si}_3\text{O}_{10} \cdot 2\text{H}_2\text{O}$)



C) Milky quartz (SiO_2)

Fig. 5.7 : Minerals with different chemical composition showing white colour

Do you know?

V^{+3} in grossular garnet (variety of garnet) causes the green colour.

Cr^{+3} causes green colour in emerald (variety of beryl)

Mn^{+3} causes red colour in muscovite mica

Cu^{+2} usually produces blue and green colours in minerals such as azurite and malachite.

2) Streak : Streak refers to the colour of a mineral's powder left behind after it is rubbed across a porcelain streak plate. This property is studied only for coloured (not white) minerals. Minerals with hardness less than 6.5 are tested with the help of the porcelain plate, while those with higher hardness are checked with other streak plates, e.g. corundum plate. Streak of a mineral may be different from the colour of the mineral e.g. pyrite, which shows golden colour has a pencil lead grey streak (Fig. 5.8).

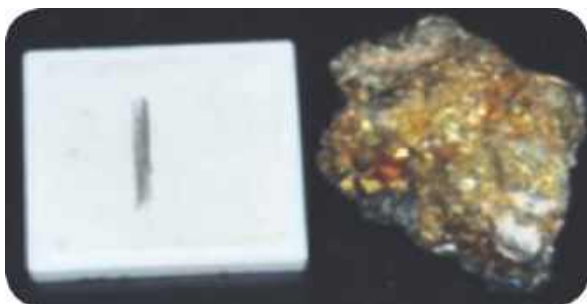
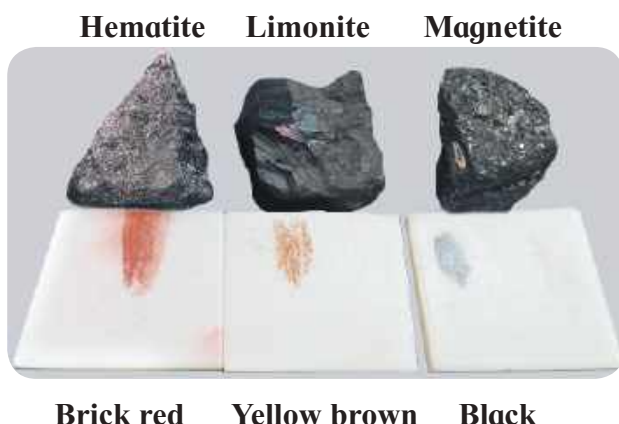


Fig. 5.8 : The streak of golden coloured mineral pyrite is pencil lead grey



Brick red Yellow brown Black

Fig. 5.9 : Streak of different dark coloured Fe-bearing minerals appears brick red, yellowish brown and black

Streak of the mineral is a more reliable identification characteristic than the colour of the mineral.

Do you know?



Application of streak in identifying gold

Kasauti stone/touch stone is a small piece of fine grained black stone which is used by goldsmiths for testing the purity of gold.

A piece of gold to be tested is scratched on the 'Kasauti' stone, which leaves a streak on it. This is compared with the streak obtained from piece of gold of known purity.

3) Lustre : Lustre describes the appearance of a mineral when light is reflected from its surface. It is the degree of reflected light and directly related to optical properties (mainly refractive index) and surface conditions.

Lustre of minerals is mainly classified as Metallic and Non-metallic :

i) Metallic lustre : Strictly belongs to opaque minerals, where light is completely reflected from the surface. Most ore minerals having high content of metals exhibit metallic lustre, e.g., galena, magnetite, pyrite (Fig. 5.10), etc.



Fig. 5.10 : Metallic lustre in pyrite

Submetallic lustre : Some minerals have lustre similar to that of a metal, but are dull and less reflective. Submetallic lustre often occurs in near-opaque minerals with very high refractive indices, such as hematite, chromite (Fig. 5.11) etc.



Fig. 5.11 : Sub-metallic lustre in Chromite

ii) Non-metallic lustres : Other lustre types are collectively known as non-metallic lustres. They may be brilliant or faint where reflection is poor due to scattering of light from the mineral surface. The different types of non-metallic lustres are:

a) Adamantine lustre : An exceptionally brilliant lustre exhibited by minerals having very high refractive index, e.g. diamond (Fig. 5.12), zircon, corundum (ruby, sapphire). These minerals are used as precious gem stones.



Fig. 5.12 Adamantine lustre in Diamond

b) Dull / Earthy lustre : Dull lustre is also known as 'earthy' and is used to describe minerals that have poor reflectivity. The surface of minerals with dull lustre is coarse and porous, e.g., kaolinite (Fig. 5.13), limonite, bauxite.



Fig. 5.13 : Dull / Earthy lustre in Kaolinite

c) Greasy / Waxy / Oily lustre : It results from light scattered by a microscopically rough surface, e.g. chalcedony (Fig. 5.14), opal etc.



Fig. 5.14 : Waxy lustre in Chalcedony

d) Pearly lustre : It results due to reflection from successive layers, such as cleavage surfaces, e.g. talc (Fig. 5.15), muscovite, gypsum.



Fig. 5.15 : Pearly lustre in Talc

e) Resinous lustre : Minerals have the appearance of resin, chewing gum or (smooth-surfaced) plastic, eg., amber (Fig. 5.16) which is a form of fossilised resin.



Fig. 5.16 : Resinous lustre in Amber

f) Silky lustre : It is due to the reflection from minerals having parallel arrangement of extremely fine fibres, e.g. gypsum (satin spar) (Fig. 5.17), asbestos, malachite.



Fig. 5.17 : Silky lustre in Gypsum (Satin spar)

g) Vitreous lustre : It is generally shown by broken glass. This type of lustre is one of the most commonly observed lustres and occurs in transparent or translucent minerals with relatively low refractive indices. Common examples include quartz (Fig. 5.18), fluorite etc.



Fig. 5.18 : Vitreous lustre in Quartz

Subvitreous lustre : When the minerals exhibit feebly developed vitreous lustre, they are said to

show subvitreous lustre, e.g. milky quartz, calcite (Fig. 5.19), etc.



Fig. 5.19 : Subvitreous lustre in Calcite

5.5.2 Characters depending upon crystal aggregate habits :

4) Form : Most minerals are formed within the spaces in between other minerals and may grow into shapeless masses. However, if they are formed freely in a hole or a cavity within a rock, the mineral takes the form of a crystal. These crystal-lined cavities are called geodes, vugs or pockets.

Form is the natural mode of occurrence of a mineral or the common geometric morphology of the mineral. Various forms are identified as follows :

Degree of crystallisation :

i) Crystallised : The minerals occur in the form of well developed crystals, e.g. quartz (Fig. 5.20).



Fig. 5.20 : Crystallised form in Quartz

- ii) **Crystalline** : The minerals occur in the form of aggregates of imperfectly developed crystals, e.g. quartz (Fig. 5.21).



Fig. 5.21 : Crystalline form in Quartz

- iii) **Cryptocrystalline** : The minerals occur as partly crystalline and partly amorphous and lack a perfect shape, e.g. agate, chalcedony (Fig. 5.22).



Fig. 5.22 : Cryptocrystalline form in Chalcedony

- iv) **Amorphous** : The minerals are formed in nature as solids without development of internal atomic structure, e.g. opal (Fig. 5.23)



Fig. 5.23 : Amorphous form in Opal

Growth dominantly in one direction :

Minerals of this category exhibit slender forms and are described as :

- i) **Acicular** : The minerals are needle-like and may occur individually or in aggregates, e.g. actinolite (Fig. 5.24).



Fig. 5.24 : Acicular form in green coloured needles of Actinolite

- ii) **Columnar** : Such minerals occur as columns of varying sizes e.g., beryl (Fig. 5.25).



Fig. 5.25 : Columnar form in Beryl (Aquamarine)

- iii) **Fibrous** : Minerals are made up of thin hair-like separable fibres or crystals, e.g. asbestos (Fig. 5.26).



Fig. 5.26 : Fibrous form in Asbestos (Serpentine)

- iv) **Radiating or Divergent** : The minerals appear as diverging needles from central nucleus e.g. natrolite (Fig. 5.27).



Fig. 5.27 : Radiating form in Natrolite

- v) **Wiry** : Some minerals occur in nature in the form of thin wires, which may be twisted or stretched in different directions, e.g. native copper or native silver (Fig. 5.28).



Fig. 5.28 : Wiry form in Native Silver

Growth in two dimensions :

Minerals of this category are described as :

- i) **Bladed** : The mineral appears like a flat blade of a knife, e.g. kyanite (Fig. 5.29).



Fig. 5.29 : Bladed form in Kyanite

- ii) **Foliated** : Mineral occurs as flakes or plates which are separable and can split into sheets or lamellae, e.g., muscovite and biotite (Fig. 5.30).



Fig. 5.30 : Foliated form in Biotite

- iii) **Tabular** : Mineral occurs in the form of a flattened rectangular shape, resembling a table, e.g. barite (Fig. 5.31), plagioclase etc.



Fig. 5.31 : Tabular form in Barite

Growth in three dimensions :

- i) **Equidimensional** : Minerals grow equally in three dimensions and appear as stubby crystals, e.g. garnet, pyrite (Fig. 5.32).

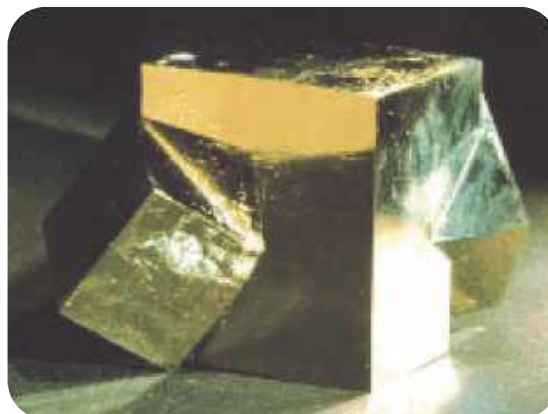


Fig. 5.32 : Equidimensional form in Pyrite

Growth with curved surfaces :

- i) **Botryoidal** : Crystals occur in spheroid aggregates resembling a bunch of grapes, e.g., chalcedony, agate (Fig. 5.33).



Fig. 5.33 : Botryoidal form in Agate

- ii) **Granular** : Mineral occurs as distinct grains resembling a lump of sugar, e.g., olivine (Fig. 5.34).



Fig. 5.34 : Granular form in Olivine

- iii) **Oolitic** : These minerals appear like small spherical growths, resembling eggs of fish, e.g. bauxite (Fig. 5.35).



Fig. 5.35 : Oolitic form in Bauxite

- iv) **Pisolitic** : Mineral grains resemble pea-sized spheroids, coarser than oolites, e.g., bauxite (Fig. 5.36).



Fig. 5.36 : Pisolitic Bauxite

- v) **Reniform** : Mineral is circular or roughly circular, resembling the shape of a kidney, e.g., hematite (Fig. 5.37), psilomelane etc.



Fig. 5.37 : Reniform form in Hematite

Growth in miscellaneous manner :

- i) **Dendritic** : Minerals show branch-like growth in different directions, e.g. native copper (Fig. 5.38).

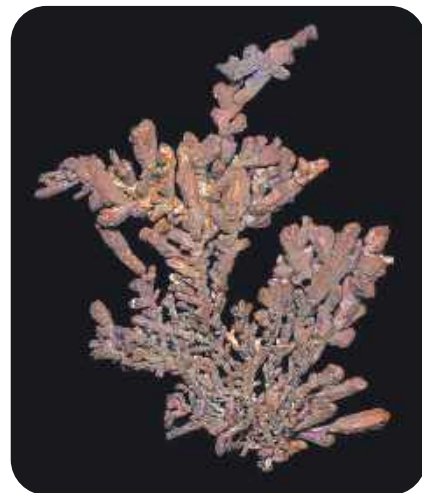


Fig. 5.38 : Dendritic form in Native Copper

- ii) **Massive / Compact** : The minerals cannot be individually identified with the naked eye as they form a compact mass exhibiting an irregular shape, e.g. rosy quartz (Fig. 5.39).



Fig. 5.39 : Massive form in Rosy Quartz

5.5.3 Characters depending upon cohesion and elasticity :

Cohesion : It is the force of attraction existing between molecules. It shows resistance to any external influence that tends to separate them, Cohesion force is related to bonding force.

Elasticity : It is the force that tends to restore the molecules of a body into their original position from which they have been disturbed.

The result of cohesion and elasticity in a mineral appears as cleavage, fracture and hardness.

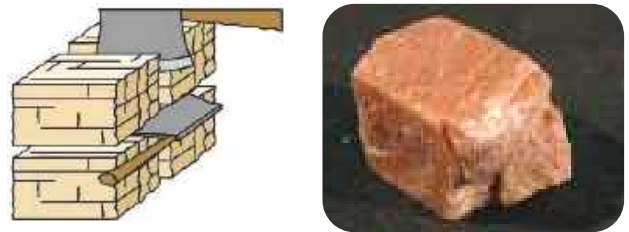
5) Cleavage : The manner in which a mineral breaks is determined by the arrangement of its atoms and the strength of the chemical bonds holding them together. As this property is unique to the mineral, careful observation can aid in mineral identification (Fig. 5.40).

The word cleavage is derived from 'cleave' meaning to break or split. Cleavage is therefore, a plane of weakness along which mineral splits easily. Only crystalline minerals can show cleavage, though every crystalline mineral may not possess cleavage (e.g. quartz). Amorphous minerals (limonite, opal etc.) do not show cleavage. Cleavage, if present, occurs as numerous parallel planes along which the

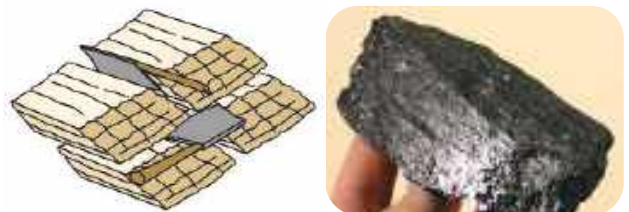
mineral is equally weak. Hence, all such weaker planes along a single direction are referred to as 'sets of cleavage'. Depending upon their atomic structure, crystalline minerals will exhibit one set of cleavage (e.g. mica, Fig. 5.40 A), two sets of cleavage (e.g. feldspars Fig. 5.40 B, amphiboles Fig. 5.40 C), three sets of cleavage (e.g. galena Fig. 5.40 D, calcite, Fig. 5.40 E) or four sets of cleavage (e.g. fluorite, Fig. 5.40 F).



A) Biotite and muscovite micas exhibit one set of cleavage.



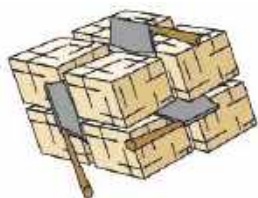
B) Orthoclase and plagioclase feldspars exhibit two sets of cleavage at approximately 90° from each other.



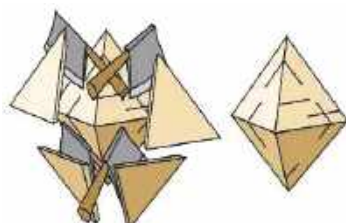
C) Hornblende (an amphibole) has two sets of cleavage at 124° . from each other.



D) Galena has three sets of cleavage that form two 90° angles (cubic cleavage)



E) Calcite has three sets of cleavage - 105° in one plane and 75° in another



F) Fluorite has four sets of cleavage (octahedral cleavage)

Fig. 5.40 : Minerals and their cleavages

Quality of cleavage can be categorised into :

- **Perfect** : Minerals with perfect cleavage split without leaving any rough surface. A full, smooth plane is formed where the crystal breaks, e.g. calcite.
- **Good** : Minerals with good cleavage also leave smooth surfaces, but often leave minor residual rough surfaces e.g. feldspars.
- **Poor** : Smooth crystal edge is not visible in minerals with poor cleavage, since the rough surface is dominant, e.g. barite and apatite.
- **Indistinct** : If a mineral exhibits cleavage, but it is so poor that it can hardly be distinguished, it is indistinct cleavage e.g. beryl.
- **None** : Some minerals never exhibit any cleavage. Thus, broken surfaces are fractured and are rough, e.g. quartz.

Do you know?

Six cleavage planes can give a 12-sided form e.g., sphalerite.

Number of cleavages and their directions	Name and description of how the mineral breaks
6 cleavages intersect at 60° and 120°	Dodecahedral cleavage Shapes made of dodecahedrons and parts of dodecahedrons

Shape of broken pieces (cleavage directions are numbered)	Illustration of cleavage directions
Sphalerite 	

6) Fracture : The appearance of broken surface of the mineral is expressed by the term fracture. It is the irregular breaking of a mineral along a surface which is not parallel to the cleavage or to the parting planes. The fracture planes are not related to the atomic structure of the mineral. These are simply ruptures which are devoid of any systematic and regular placement of planes. The different types of fractures are described as follows :

- i) **Conchoidal** : Broken surface shows concentric rings or concavities which may be deep in character, e.g., obsidian (Fig. 5.41 A). When the broken surface shows faint concentric rings, it is subconchoidal fracture. e.g. rock crystal, flint (Fig. 5.41 B)

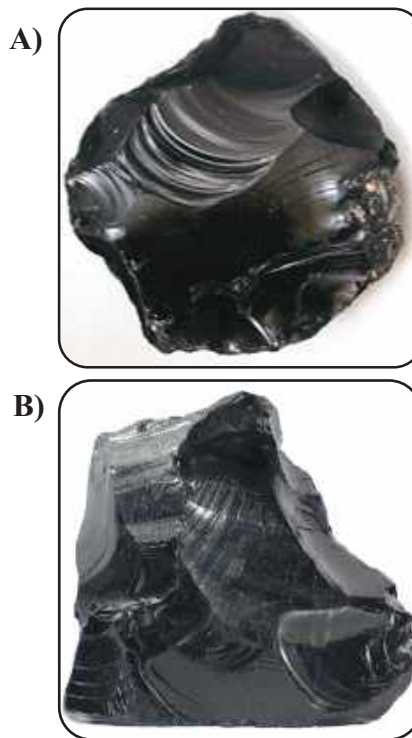


Fig. 5.41 : A) Conchoidal, B) Subconchoidal fracture

- ii) **Earthy** : Fractured surface is soft and smooth resembling the broken surface of chalk, e.g. bauxite, limonite (Fig. 5.42).



Fig. 5.42 : Earthy fracture in Limonite

- iii) **Hackly** : Broken surface is highly irregular and exhibits sharp, pinching projections. e.g. asbestos, native copper (Fig. 5.43).



Fig. 5.43 : Hackly fracture in Native Copper

- iv) **Splintery** : The mineral breaks with a rough woody fracture, e.g. kyanite, serpentine (Fig. 5.44).



Fig. 5.44 : Splintery fracture in Serpentine

- v) **Uneven** : Broken surface is rough and irregular with a rugged appearance, e.g.,

orthoclase, magnetite (Fig. 5.45).



Fig. 5.45 : Uneven fracture in Magnetite

7) Hardness : Hardness of mineral is a measure of its resistance to abrasion or scratching.

The degree of hardness is determined by observing the comparative ease or difficulty with which one mineral is scratched by another. Hardness of a mineral depends upon its atomic structure, hence it is considered to be a reliable property in mineral identification.

In 1822, an Austrian mineralogist Friedrich Mohs devised a hardness scale, (Table 5.1) based on one mineral's ability to scratch another mineral. He placed 10 minerals in order, from softest to hardest, giving a relative hardness value of 1 to the softest mineral i.e. talc and 10 to the hardest i.e. diamond. Each mineral on the scale scratches the one below it (the lower number), but not the one above it (the higher number). This is known as Mohs' scale of hardness, and is one of the best practical methods of estimating a mineral's hardness :

Note : It must be noted that Mohs' scale is arbitrary and non-linear (Fig. 5.46), i.e. the steps between relative hardness values are not necessarily equal. Rather, it is a method of gauging the relative hardness of minerals. If a mineral cannot be scratched by a knife blade but can be scratched by quartz then its hardness is between 5 and 7 (stated as 5-7) on Mohs' scale. A relative hardness value of 6.5 means, that the mineral could scratch orthoclase (feldspar) but not quartz.

Table 5.1 : Mohs' hardness scale with objects of equivalent hardness

	Mineral Name	Scale Number	Common Object
Increasing Hardness	Diamond	10	Masonry Drill Bit (8.5)
	Corundum	9	
	Topaz	8	
	Quartz	7	
	Orthoclase	6	Steel Nail (6.5)
	Apatite	5	Knife/Glass Plate (5.5)
	Fluorite	4	Copper Penny (3.5)
	Calcite	3	
	Gypsum	2	
	Talc	1	Finger nail (2.5)

Test : Using moderate pressure, drag a sharp edge over the smooth surface of a mineral. If the surface of the mineral is scratched, then it is softer than the material used to scratch it, if not then it is harder.

Terminology :

Soft : can be scratched by a fingernail (1-2 Mohs').

Medium : can be scratched by a knife or nail (3-5 Mohs').

Hard : cannot be scratched by a knife, but it can scratch glass (6-9 Mohs').

Diamond is the hardest known mineral (10 Mohs').

Do you know?

Hardness is measure of the strength of the structure of the mineral relative to the strength of its chemical bonds. It is not the same as brittleness, which is another

measure of strength that is purely related to the structure of the mineral. Minerals with small atoms, packed tightly together with strong covalent bonds throughout, tend to be hardest minerals. The soft minerals have metallic bonds or even weaker Van der Waals bonds as important components of their structure. Hardness is generally consistent with the chemistry of minerals.

Minerals exhibiting hardness more than 7 can be cut and polished as gemstones, e.g. crystalline varieties of quartz, corundum, beryl, etc. Any mineral with hardness less than 7 cannot be used as a gemstone, but can be used in making sculptures or artifacts, e.g. jade, chalcedony, malachite, turquoise, etc.

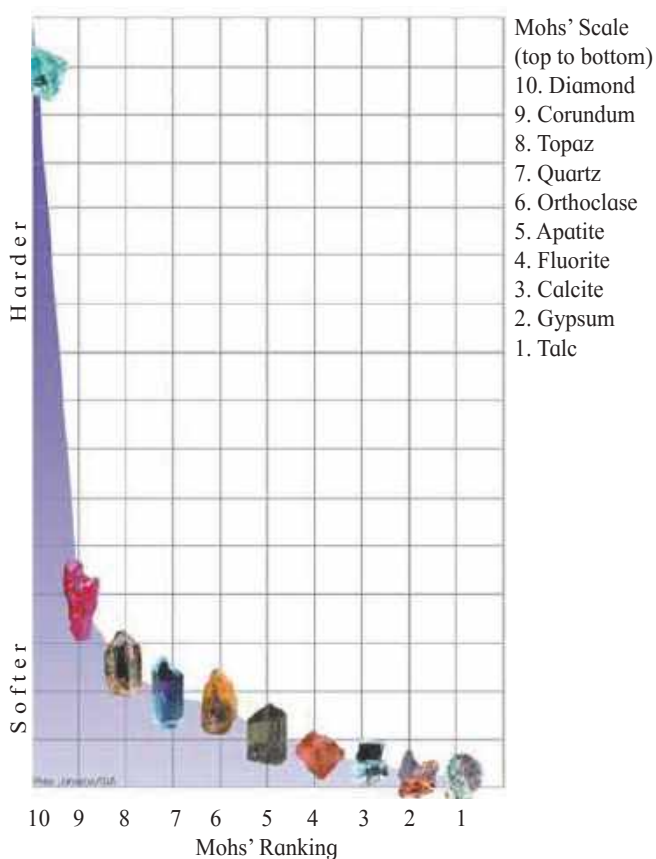


Fig. 5.46 : Non-linear relationship between Mohs' scale and hardness of minerals

8) Specific Gravity : Specific gravity of a mineral is defined as a number that expresses the ratio of the weight of the mineral in air to the weight of an

equal volume of water displaced by it at normal temperature and pressure (NTP) conditions. The specific gravity of a mineral depends upon the atomic arrangement, crystal structure, chemical composition. It varies with temperature and pressure. For a mineral of variable composition crystallising in specific structure, the variation in specific gravity will depend on the atomic weight of the individual atoms, e.g., in olivine, specific gravity varies from 3.2 for pure forsterite (Mg_2SiO_4) to 4.4 for pure fayalite (Fe_2SiO_4). The atomic weight of Mg is 24.3 and Fe is 55.8.

The common rock forming minerals have specific gravity that ranges between 2.5 and 3.5 while the average specific gravity for metallic minerals is about 5.0. Ore minerals have specific gravity around 3.0.

The specific gravity of minerals can be determined in the laboratory by two methods i.e. Walker's steelyard balance and Jolly's spring balance.

Do you know?

Minerals like galena (7.5), barite (4.25) and hematite (5.26) having high specific gravity are commonly used to increase the weight of drilling muds for oil- well drilling especially to avoid blow-out. Gravity separation technique depends on the difference in specific gravity of minerals. This method is environment friendly and used for the separation of high specific gravity minerals and metallurgical waste.

Classification of Minerals :

On the basis of chemical composition and internal structure, minerals are classified as follows :

Oxides : Oxides are minerals that consist of metal cations (atoms with a positive charge) bonded to oxygen anions. An anion is an atomic structure that has a net negative charge, e.g. haematite (Fe_2O_3) and magnetite (Fe_3O_4).

Sulphides : Sulphides consist of metal cations bonded to a sulphide anion, e.g. pyrite (FeS_2) and galena (PbS).

Sulphates : Sulphates are minerals that consist of metal cations bonded to a sulphate (SO_4^{-2}) anion. e.g. gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$).

Halides : Halides are minerals that consist of metal cations bonded to halide anions. Halide refers to the elements chlorine, fluorine, iodine, e.g. fluorite (CaF_2).

Carbonates : Carbonates are minerals that contain metal cations bonded to the carbonate (CO_3^{-2}) anion, e.g. calcite (CaCO_3).

Native Elements : Minerals that consist of one single element belong to this group, e.g. copper, gold and silver.

Silicates : Silicates make up over 95% of the Earth's crust and about 97% of the mantle. They are the most common minerals on Earth. Silicates are minerals that contain the fundamental silicon tetrahedra (SiO_4) (Fig. 5.47). A tetrahedra is a geometric shape similar to a four-sided pyramid. The silica tetrahedra is composed of one silicon atom (Si^{+4}) surrounded by four oxygen atoms (O^{-2}).

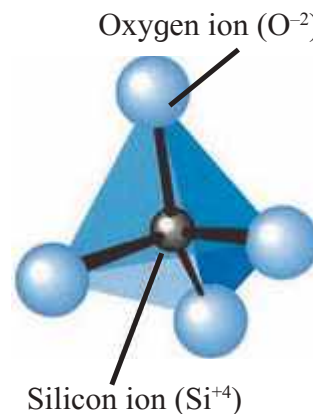


Fig. 5.47 : Silicon tetrahedra

Classification based on structure of silicates :

SiO_4 tetrahedra is the fundamental unit of all silicate minerals. This unit can either occur alone or in different combinations (along with other cations) in the mineral structure. Different groups of silicate structures of rock forming minerals are as follows :

- i) **Nesosilicates** : In this group, SiO_4 tetrahedra occur as independent units, (Fig. 5.48) e.g., olivine, garnet, kyanite etc.

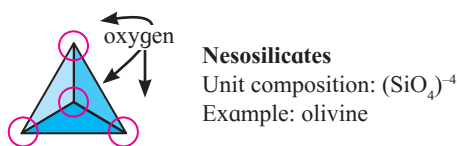


Fig. 5.48 : Nesosilicates

- ii) **Sorosilicates** : SiO_4 tetrahedra occur in pairs, (Fig. 5.49) e.g., epidote etc.

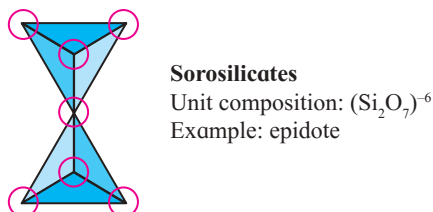


Fig. 5.49 : Sorosilicates

- iii) **Inosilicates** : These are also called as 'chain silicates'. In this group of minerals SiO_4 tetrahedra occur as chains resulting in preferential growth of mineral along one direction. Two varieties of Inosilicates occur in nature. They are single chain silicates (Fig. 5.50 A) e.g. pyroxenes and double chain silicates (Fig. 5.50 B) e.g. amphiboles.

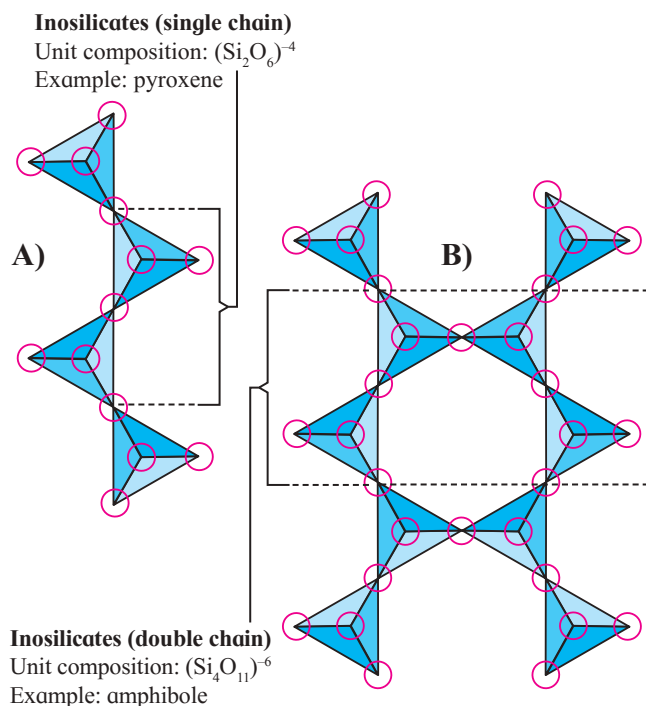


Fig. 5.50 : Inosilicates

- iv) **Cyclosilicates** : These are known as 'ring silicates'. In this group of minerals, 3 or 4 or 6 SiO_4 tetrahedra occur in the form of a ring (Fig. 5.51) e.g., beryl, tourmaline.

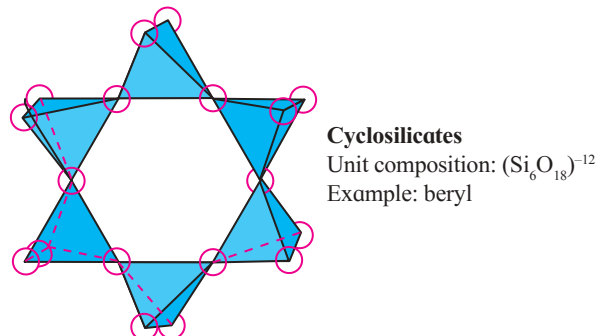


Fig. 5.51 : Cyclosilicates

- v) **Phyllosilicates** : These are 'sheet silicates'. In this group, SiO_4 tetrahedra occur as sheets, resulting in preferred growth along two directions of a mineral (Fig. 5.52) e.g., micas, talc, serpentine.

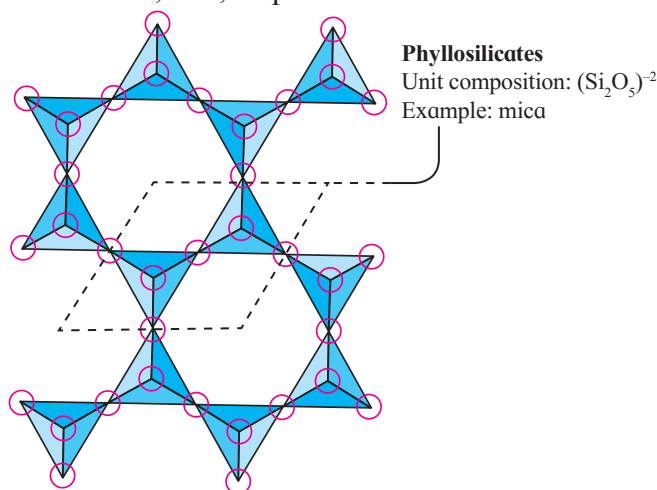


Fig. 5.52 : Cyclosilicates

- vi) **Tectosilicates** : In this group, SiO_4 tetrahedra occur in a three-dimensional framework resulting in equidimensional growth of minerals (Fig. 5.53) e.g., quartz, feldspars, zeolites.

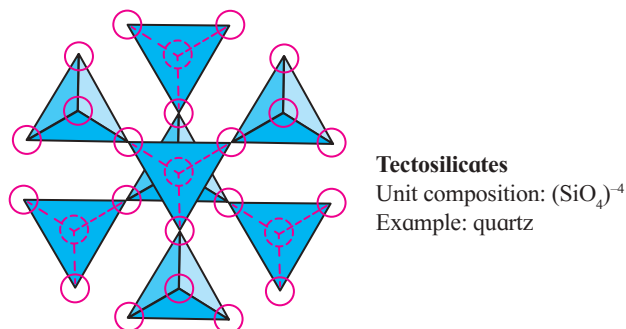


Fig. 5.53 : Tectosilicates

Rock forming mineral groups :

Rocks are composed of minerals. Only a few of the minerals are rock forming. Most rocks however, are a combination of commonly occurring minerals such as feldspars, quartz, micas, olivine, pyroxene and amphiboles.

Feldspar group :

Feldspar is one of the most abundant mineral found in the continental crust. This group constitutes over 50% of the Earth's crust.

The name feldspar is derived from the German word 'Feldspat'. 'Feld' meaning field and 'Spat' meaning a rock (that does not contain ore).

Silicate structure : All feldspars are tectosilicates. Ca, Na, K cations also occur in tectosilicates.

Chemical composition : It is expressed as $X(\text{AlSiO})_4\text{O}_8$, where X is Ca, Na, K and Ba.

Varieties : This group is sub-divided on the basis of chemical composition and isomorphism into :

i) Alkali feldspars ii) Calc-alkali feldspars

i) Alkali feldspars or Potassium Sodium feldspars. e.g. orthoclase and microcline.

Albite ($\text{NaAlSi}_3\text{O}_8$) to Orthoclase

(KAlSi_3O_8) isomorphous minerals. It is a substitution of sodium by potassium.

Orthoclase : Name has its origin in the Greek words- 'Orthos' meaning right, 'Clase' meaning to cleave (cleavages are at right angles) (Fig. 5.54).



Fig. 5.54 : Orthoclase

Microcline : 'Micro' meaning small or little, 'Cline' meaning to incline or to lean (Fig. 5.55).

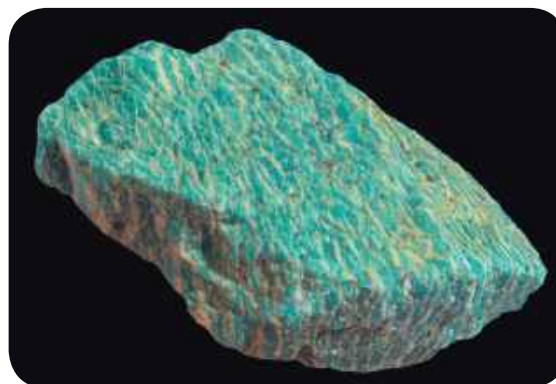


Fig. 5.55 : Microcline

Table 5.2 : Varieties of feldspar group of minerals can be given as follows:

Alkali feldspar	Potash feldspar (KAlSi_3O_8)	Barium feldspar ($\text{BaAl}_2\text{Si}_2\text{O}_6$)
	Orthoclase	Hyalophane Celsian.
	Microcline	
	Soda Orthoclase	
	Soda Microcline	
Alkali feldspar	Albite ----- Oligoclase --- Andesine --- Labradorite --- Bytownite --- Anorthite. ($\text{NaAlSi}_3\text{O}_8$)	($\text{CaAl}_2\text{Si}_2\text{O}_8$)
	Soda feldspar	Lime feldspar
	Plagioclase or Lime soda feldspar	

- ii) Calc-alkali feldspars - Plagioclase series of feldspars or lime soda feldspars, e.g. plagioclase (Fig. 5.56) and labradorite (Fig. 5.57).



Fig. 5.56 : Plagioclase



Fig. 5.57 : Labradorite

Plagioclase feldspars : Plagioclase series of feldspar minerals are a homogeneous mixture of albite and anorthite. In this series Na-Si is replaced by Ca-Al. The minerals so formed are called isomorphs. The two boundary minerals albite and anorthite are called end members.

Albite (Ab) $\rightleftharpoons$ Anorthite (An).

Table 5.3 : Plagioclase minerals and their composition :

Names	Ab content (%)	An content (%)
1. Albite	100-90	0-10
2. Oligoclase	90-70	10-30
3. Andesine	70-50	30-50
4. Labradorite	50-30	50-70
5. Bytownite	30-10	70-90
6. Anorthite	10-0	90-100

Physical properties of feldspar group :

Colour : Blue, orange, pink, white, green and gray.

Streak : White or pale shade of body colour.

Lustre : Vitreous to subvitreous

Form : Tabular

Cleavage : Two sets, at right angles

Fracture : Conchoidal to uneven

Hardness : 6 - 6.5

Specific gravity : 2.55 - 2.76

Occurrence : Feldspars are found in igneous rocks (e.g., granite, syenite, pegmatite), sedimentary rocks (e.g., arkose a variety of sandstone containing at least 25% feldspar) and metamorphic rocks (e.g., gneiss).

Uses : Feldspars are used in the manufacture of glass, porcelain, sanitary ware and as filler in paints, plastic, rubber and adhesive industries. Feldspars are also used as gemstones e.g., orthoclase as moonstone, labradorite as sunstone and microcline as amazonite.

Silica group :

Quartz or 'Gargoti' was used during the stone age. The word quartz is derived from the German word 'Quarz' which came from the Polish dialect term 'Kwardy', corresponding to the Czech term 'Tvrdy' meaning hard. Quartz is the second most abundant mineral in the Earth's crust.

Silica group constitutes 12% to 15% of all rock forming minerals. This group includes crystalline, cryptocrystalline and amorphous minerals.

Silicate structure : Silica minerals are tectosilicates.

Chemical composition : The minerals of this group are characterised by same chemical composition i.e. SiO_2 . Crystalline quartz is most abundant.

The composition varies from crystalline to cryptocrystalline silica, i.e. from SiO_2 to $\text{SiO}_2 \cdot n\text{H}_2\text{O}$. Varieties of cryptocrystalline silica are mixtures of cryptocrystalline silica (chalcedony) and hydrous silica (opal).

Varieties : Several varieties of silica can be recognised on the basis of their forms :

1) Crystalline silica (SiO_2) :

Well defined crystals are absent, but a marked tendency towards crystallisation is exhibited.

Rock crystal : Colourless and transparent (Fig. 5.58).



Fig. 5.58 : Rock crystal

Amethyst : Violet or purple coloured (Fig. 5.59), colour resulting from trace amounts of Fe impurity.



Fig. 5.59 : Amethyst

Rosy quartz : Pink colour (Fig. 5.60), resulting from traces of Ti impurity.



Fig. 5.60 : Rosy quartz

Milky quartz : Milky white coloured (Fig. 5.61), colour due to numerous air cavities.



Fig. 5.61 : Milky quartz

2) Cryptocrystalline silica ($\text{SiO}_2 + \text{SiO}_2 \cdot n\text{H}_2\text{O}$) :

Crystals are partially developed. Some faces of the crystals are developed and remaining faces are not (amorphous).

Chalcedony : A fibrous variety of quartz, (Fig. 5.62), white, brown or gray in colour, shows waxy lustre and is found as cavity fillings. Name derived from the town of 'Chalcedon', a district in Istanbul, Turkey.



Fig. 5.62 : Chalcedony

Agate : A variety of chalcedony, exhibits alternate curving layers of different colours (Fig. 5.63). Name is derived from the occurrence of the stone along the shoreline of river Achates (now called Dirillo), in south western Sicily.



Fig. 5.63 : Agate

Onyx : Also a variety of chalcedony, exhibits alternate, parallel white and black or dark brown layers (Fig. 5.64).



Fig. 5.64 : Onyx

Jasper : An impure, opaque variety, red, brown, (Fig. 5.65) or yellow in colour. Name has its origin in the Latin word 'Gasper' from the Biblical Hebrew word 'Gizbar' meaning treasurer.



Fig. 5.65 : Jasper

Flint : Usually black or in shades of gray, found as nodules and breaks with sharp edges (Fig. 5.66). Name has been derived from Proto-Indo-European word 'Splind' meaning to split or cleave.



Fig. 5.66 : Flint

3) Amorphous silica ($\text{SiO}_2 \cdot n\text{H}_2\text{O}$) :

There is total absence of crystallinity, not a single crystal face developed.

Opal : An amorphous variety of silica. Name derived from Sanskrit word 'Upala' meaning precious stone (Fig. 5.67).



Fig. 5.67 : Opal

Physical properties of silica group :

Colour : Colourless or white, also shows variety of colours.

Streak : White.

Lustre : Vitreous, subvitreous, waxy.

Form : Prismatic, botryoidal, massive.

Cleavage : Absent.

Fracture : Conchoidal to subconchoidal, uneven.

Hardness : 5.5 - 7 (variable depending upon silica variety)

Specific gravity : about 2.65

Occurrence : It occurs in igneous rocks like granite, pegmatite; as secondary mineral within cavities of basalts; in sedimentary rocks such as sandstone and in metamorphic rocks like quartzite, schist and gneiss.

Uses : In glass, ceramics, abrasive and refractory industries, piezoelectric crystal plates used in quartz watches and also used as decorative and semi-precious stone.

Flint was used in the manufacture of tools as it has the property of splitting into thin, sharp splinters when struck by another hard object.

Amphibole group :

The name 'Amphibole' is derived from Greek word 'Amphibolus' meaning ambiguous.

Amphibole is an important group of rock forming minerals which are basically dark coloured (mafic).

Silicate structure : Amphiboles are double-chain inosilicates.

Chemical composition : These minerals are complex silicates of Mg, Fe, Ca and Al with a hydroxyl group (OH radical) present.

Varieties : Amphiboles include hornblende, tremolite and actinolite.

Hornblende : Hornblende is the most common mineral. Name is derived from German terms 'Horn' referring to its colour and 'Blende' meaning to deceive, owing to its similarity in appearance to metal-bearing ore minerals (Fig. 5.68).



Fig. 5.68 : Hornblende

Tremolite : Name has been derived from Tremola Valley, Switzerland (Fig. 5.69).



Fig. 5.69 : Tremolite

Actinolite : Name has been derived from the Greek words 'Aktim' for ray and 'Lithos' for stone, meaning fibrous in nature (Fig. 5.70).



Fig. 5.70 : Actinolite

Physical properties of amphibole group :

Colour : Black, dark green, dark brown

Streak : White, colourless.

Lustre : Vitreous, silky, dull.

Form : Prismatic, acicular, fibrous, bladed, columnar and massive

Cleavage : 2 sets, prismatic, at 124° and 56° .

Fracture : Uneven.

Hardness : 5 - 6

Specific gravity : 2.9 - 3.4

Occurrence : Hornblende is an important constituent of many igneous rocks such as granite, syenite, diorite and andesite. Amphiboles are common in metamorphic rocks e.g. schist and gneiss.

Uses : ‘Nephrite’, a variety of amphibole is fibrous aggregate of actinolite and tremolite. It is used as a gemstone, in sculptures, jewellery, tools, etc.

Pyroxene group :

The name ‘Pyroxene’ has its origin in the Greek words- ‘Pyro’ meaning fire and ‘Xenox’ meaning stranger.

Pyroxene is an important rock forming mineral group, which is similar to amphibole group, except for different atomic structure and the absence of hydroxyl group (OH radical).

Silicate structure : All pyroxenes are single chain inosilicates.

Chemical composition : Pyroxenes are complex silicates of Mg, Fe, Ca and Al.

Varieties : Pyroxenes are a group of allied minerals. The common members of this group are augite and hypersthene.

Augite : $(\text{Ca, Na, Mg, Fe, Mn, Ti, Al})_2(\text{Al, Si})_2\text{O}_6$ is a common brownish black variety.

Name has been derived from the Greek word ‘Auge’ meaning shine or lustre, in allusion to the appearance of its cleavage surface (Fig. 5.71).



Fig. 5.71 : Augite

Hypersthene : $(\text{Mg, Fe})\text{SiO}_3$

It is gray or greenish black in colour with metallic lustre.

Name has its origin from the Greek words ‘Hyper’ meaning exceeding, ‘Sthenos’ meaning strength (Fig. 5.72).

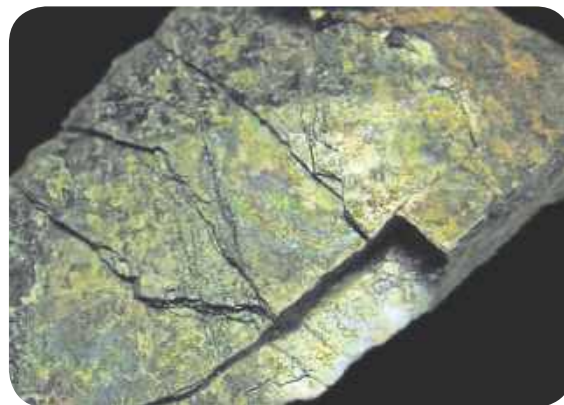


Fig. 5.72 : Hypersthene

Physical properties of pyroxene group :

Colour : Gray, greenish brown, brownish black.

Streak : White or pale body colour

Lustre : Vitreous, subvitreous.

Form : Prismatic, granular, massive.

Cleavage : Two sets, at nearly 90° .

Fracture : Uneven

Hardness : 5 - 6

Specific gravity : 3.22 - 3.56 (increases with Fe content)

Occurrence : Pyroxenes are found in basic and ultrabasic igneous rocks like gabbro, dolerite, basalt and peridotite.

Augite is found in metamorphic rocks like schist and gneiss and hypersthene is found in charnockites and granulites.

Uses : Used in ceramics. Jadeite is used as a gemstone.

Do you know?

Nephrite (an amphibole) and Jadeite (a pyroxene) are two different minerals which together form mineral Jade.

Mica group :

Micas are important rock forming minerals. They can be cleaved into thin elastic plates.

Name Mica has its origin in the Latin word ‘Micare’ meaning to flash or glisten.

Silicate structure : All micas are phyllosilicates.

Chemical composition : Micas are silicates of Al and K with Mg or Fe. Some varieties contain Na, Li or Ti. Hydroxyl group is present in micas and is partially replaced by fluorine.

Varieties :

Muscovite : Hydrated silicate of Aluminium and Potassium with Fluorine, silvery white in colour. Name Muscovite originated from 'Muscovy glass' because it came from Muscovy province of Russia (Fig. 5.73).



Fig. 5.73 : Muscovite

Biotite : Hydrated silicate of Mg, Fe, Al, and K with F, black to brown in colour (Fig. 5.74).



Fig. 5.74 : Biotite

Physical properties of mica group :

Colour : Colourless, silvery white, dark green, brown, black.

Streak : White, colourless.

Lustre : Pearly, silky.

Form : Foliated, flaky, lamellar

Cleavage : One set, perfect

Fracture : Uneven.

Hardness : 2 - 3

Specific gravity : 2.76 - 3.1

Occurrence : In igneous rocks, it occurs in pegmatite and granite; in sedimentary rocks such as sandstone and in metamorphic rocks like schist and gneiss.

Uses : As insulator in electrical industries, as filler in rubber, in lubricants and paints, has wide applications in ayurvedic medicines.

Do you know?

Mica is a good electrical insulator that can withstand extremely high temperatures. It is commonly used in equipments like toasters, electric irons and hair dryers where the heating elements are sandwiched between mica sheets.

Olivine group :

Like amphiboles and pyroxenes, olivines are also mafic i.e., dark coloured minerals.

Silicate structure : Olivines are nesosilicates.

Chemical composition : Olivine is chemically $(\text{Mg,Fe})_2\text{SiO}_4$ with two end members, forsterite $(\text{Mg}_2\text{SiO}_4)$ and fayalite $(\text{Fe}_2\text{SiO}_4)$.

Varieties : The olivine group consists of an isomorphous series between forsterite and fayalite.

Forsterite : It is green, yellow, yellowish green, white.

Fayalite : It is greenish yellow, yellowish brown or brown.

Olivine : It has been named after its typical olive green colour (Fig. 5.75).



Fig. 5.75 : Olivine

Physical properties of olivine group :

Colour : Usually olive green.

Streak : White.

Lustre : Vitreous, but often shows dull lustre due to alteration by weathering.

Form : Granular, massive.

Cleavage : Absent

Fracture : Conchoidal to uneven.

Hardness : 6.5 - 7

Specific gravity : 3.2 - 4.3 (depending on Fe content)

Occurrence : It occurs in monomineralic ultrabasic igneous rocks like dunite. It also occurs in rocks like peridotite (olivine and pyroxene), basalt, dolerite and gabbro.

Uses : Olivine is used in the manufacture of refractory bricks and its variety Peridot is used as a gemstone.

Zeolite group :

The term 'Zeolite' was originally coined by Swedish mineralogist Axel Fredrick Cronstedt, who observed that rapid heating of the mineral produced large amounts of steam from water that had been adsorbed by the mineral. Based on this, he named the mineral as Zeolite, from the Greek word 'Zein' meaning to boil.

Silicate structure : Zeolites are tectosilicates.

Chemical composition : Zeolites are aluminosilicates of alkali and alkaline earth metals.

Do you know?

Zeolites belong to the family of solids known as 'molecular sieves'. Molecular sieves are crystalline metal aluminosilicates having a three dimensional interconnecting network of silica and alumina tetrahedra. Natural water of hydration is removed from this network by heating, to produce uniform cavities which selectively adsorb molecules of a specific size.

Varieties :

Heulandite : Named after John Henry Heuland, a British mineral collector and dealer (1778-1856). It shows tabular form and vitreous to pearly lustre (Fig. 5.76).



Fig. 5.76 : Heulandite

Stilbite : Originated from the Greek word 'Stilbein' meaning to glitter or shine or 'Stilbe' a mirror, alluding to its pearly or vitreous lustre. It shows unique 'bowtie' form (Fig. 5.77).



Fig. 5.77 : Stilbite

Scolecite : Named from the Greek word 'Skolex' which means worm, with reference to the mineral's reaction to a blowpipe flame. Crystals radiate out from a centre without producing stellar forms (Fig. 5.78).



Fig. 5.78 : Scolecite

Mesolite : Originated from the Greek word ‘Meso’ meaning middle and ‘Lithos’ meaning stone, due to its chemical composition being in between that of **Natrolite** and Scolecite. It shows cotton-ball cluster of acicular white crystals (Fig. 5.79).



Fig. 5.79 : Mesolite

Physical properties of zeolite group :

Colour : White, red, pink, yellow, brown.

Streak : White

Lustre : Pearly to vitreous.

Form : Crystals are usually thin and delicate, acicular or may occur as sheaf-like aggregates.

Cleavage : Perfect, one or two sets.

Fracture : Subconchoidal to uneven.

Hardness : 3.5 - 5.0

Specific gravity : 2.1 - 2.4

Occurrence : They usually occur in cavities and veins of volcanic basic rocks such as basalts (Deccan trap), also in acid igneous rocks like rhyolite. Few zeolite minerals occur in tuffaceous sediments.

Uses : Zeolites are mainly used in domestic and commercial water softening and purification.

5.8 : Rock forming minerals :

Apophyllite :

Apophyllite, according to recent studies, is the name of a group of minerals and is not a specific mineral title.

It is generally found along with zeolites and has very high water content.

The name apophyllite has been derived from Greek words ‘Apo’ meaning away from and ‘Phyll’ meaning leaf, referring to its tendency to exfoliate (i.e. flake apart like a leaf) when heated due to loss of water.

Silicate structure : Apophyllite is a phyllosilicate.

Chemical composition :



Varieties : On the basis of variable ratio of fluorine and hydroxyl group, apophyllites can be differentiated into the following mineral species :

i) **Fluorapophyllite** : White, colourless, yellow, green, (F > OH); (Fig. 5.80).



Fig. 5.80 : Fluorapophyllite

ii) **Hydroxyapophyllite** : White, colourless, (OH > F); (Fig. 5.81).



Fig. 5.81 : Hydroxyapophyllite

iii) **Natroapophyllite** : Brown, yellow, colourless, (K replaced by Na) (Fig. 5.82).



Fig. 5.82 : Natroapophyllite

Physical properties of apophyllite :

Colour : Usually white, green, yellow, also blue, brown, pink, violet.

Streak : White

Lustre : Vitreous, pearly

Form : Prismatic, tabular, massive

Cleavage : Perfect, one set.

Fracture : Uneven

Hardness : 4.5 - 5

Specific gravity : 2.3 - 2.4

Occurrence : Apophyllite is a secondary mineral which occurs as cavity fillings in vesicles of volcanic igneous rocks like basalt.

Uses : As a decorative mineral and as an ornamental stone.

Do you know?

Cavansite is also found along with zeolites and apophyllite. It is found in many regions of the world, but it is generally accepted that the ones from Wagholi Dist, Pune, Maharashtra excel in quality and quantity to those found in other countries.



A cluster of cavansite mineral crystals shown on a bed of microcrystalline stilbite, quarried in Wagholi, Pune, Maharashtra.

Talc :

Mineral talc (Fig. 5.83) is a soft, non-metallic mineral.

The name has been derived from Arabic word 'Talq', due to the pure colour of its powder.

Silicate structure : Talc is a phyllosilicate.

Chemical composition : Talc is hydrated magnesium silicate, $\text{Mg}_3\text{Si}_4\text{O}_{10}(\text{OH})_2$.

Varieties :

i) **Soapstone or Steatite (Talc schist)** : It is a metamorphic rock composed predominantly of talc which has a soft texture and soapy feel.

ii) **Pot stone** : It is impure Steatite, hard, grey or black coloured, used especially in prehistoric times to make cooking vessels.

iii) **French chalk** : It is a kind of Steatite used for marking cloth and removing grease, also used in powder form as a dry lubricant.



Fig. 5.83 : Talc

Physical properties of talc :

Colour : White, pale green, brown, gray, yellowish white, brownish white.

Streak : White

Lustre : Vitreous, pearly, greasy, waxy.

Form : Foliated, fibrous.

Cleavage : One set, perfect.

Fracture : Uneven

Hardness : 1 (Mohs' scale), very soft and easily scratched by finger nail.

Specific gravity : 2.7 - 2.8

Occurrence : Talc commonly occurs in metamorphic rocks like talc schists.

Uses : Talc is used in paint, paper, ceramics, plastics, cosmetics and talcum powders. Soapstone is used to make ornamental objects, sculptures, bowls, countertops, sinks, etc.

Do you know?

During Stone age, the people of Scandinavia used soapstone carved moulds to cast metal objects such as knife blades and spear heads.

Seals of Mohenjodaro and Harappan civilization were made from soapstone (steatite).

Gypsum :

Name 'Gypsum' originated from the Greek word 'Gypsos' meaning plaster. Mineral gypsum found in abundance near Paris is called Plaster of Paris.

Gypsum is the most common sulphate mineral found on the Earth's crust, extracted, processed and used by man in construction or decoration in the form of plaster and alabaster since 9000 BCE.

It is an indefinitely recyclable raw material which can be reused because the chemical composition of the raw material in the final product remains unchanged.

Silicate structure : Gypsum is a non-silicate mineral belonging to sulphate group i.e. it lacks silicon.

Chemical composition : Hydrated calcium sulphate, ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$).

Varieties :

i) **Alabaster gypsum :** Fine grained and off white coloured (Fig. 5.84).



Fig. 5.84 : Alabaster gypsum

ii) **Selenite gypsum :** Transparent and bladed (Fig. 5.85).



Fig. 5.85 : Selenite gypsum

iii) **Satinspar gypsum :** Pearly and fibrous (Fig. 5.86).



Fig. 5.86 : Satinspar gypsum

Physical properties of gypsum :

Colour : Colourless, milky white, grayish, yellowish, red or brown

Streak : White

Lustre : Vitreous, pearly, silky.

Form : Elongated and generally as prismatic crystals, granular, lamellar, fibrous.

Cleavage : Perfect, 2 sets.

Fracture : Hackly, brittle.

Hardness : 2 (Mohs' scale)

Specific gravity : 2.31 - 2.33

Occurrence : As an evaporite mineral and as a hydration product of anhydrite, in bedded deposits associated with limestones, shales and salt deposits, also formed by precipitation from natural waters.

Uses : In the manufacture of plaster of paris, blackboard chalk, wallboard, in cement industry to control the setting or hardening time of cement, as a fertilizer for plant growth in soils with deficiency of Ca and S.

Calcite :

Calcite is one of the most common minerals on the surface of the Earth. It is the crystalline form of CaCO_3 , occurring as chalk, limestone and marble (Fig. 5.87).

Silicate structure : It is a non-silicate mineral belonging to the carbonate group.

Chemical composition : Calcite is calcium carbonate (CaCO_3).

Varieties : Iceland spar : Pure, transparent and colourless showing double refraction (Fig. 5.88).



Fig. 5.87 : Calcite



Fig. 5.88 : Iceland spar

Physical properties of calcite :

Colour : Colourless, white, gray, black, brown, yellow, red, blue, green

Streak : White.

Lustre : Vitreous, pearly, resinous, dull.

Form : Crystalline (rhombohedral), massive

Cleavage : Perfect, 3 sets.

Fracture : Uneven, conchoidal (rarely observed due to perfect cleavage).

Hardness : 3 (Mohs' scale)

Specific gravity : 2.71 - 2.94

Occurrence : Calcite occurs in sedimentary rocks like limestone and metamorphic rocks such as marble, as vein and cavity fillings in igneous rocks like basalt and as deposits in limestone caves.

Uses : The pure variety of calcite i.e. Iceland spar is used in the manufacture of optical lenses, optical instruments and Nicol's prism. Calcite is used in cement industry, in manufacture of soap, paints, paper, abrasive, in soil treatment as a conditioner and also in medicines (Fig. 5.88).

Do you know?

We can differentiate between the minerals Gypsum and Calcite with an acid test. Calcite (CaCO_3) gives effervescence of CO_2 even when cold weak acids such as vinegar are placed on the specimen, whereas Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) shows no reaction.



Fluorite :

The name Fluorite (Fig. 5.89) was derived from Latin word 'Fluere' meaning to flow. In the mining industry, it is often called Fluorspar.

Silicate structure : Fluorite is a non-silicate mineral belonging to halide group.

Chemical composition : Calcium fluoride (CaF_2)



Fig. 5.89 : Fluorite

Physical properties of fluorite :

Colour : Colourless, deeply coloured due to impurities.

Streak : White

Lustre : Vitreous, dull

Form : Crystalline (octahedral), massive

Cleavage : Four sets, often cleaves into perfect octahedrons.

Fracture : Subconchoidal, uneven.

Hardness : 4 (Mohs' scale)

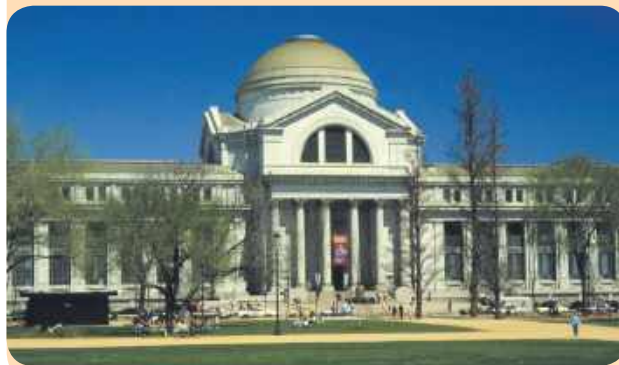
Specific gravity : 3.18 - 3.56

Occurrence : It is a very common rock forming mineral. It often occurs as a gangue mineral associated with metallic ores deposited in veins by hydrothermal processes. It is also found in fractures and cavities of some sedimentary rocks such as limestones and dolomites.

Uses : Fluorite is an important industrial mineral used in chemical, metallurgical and ceramic industries. It is also used as a gemstone and ornamental stone.

Do you know?

The Smithsonian National Museum of Natural History is the world's largest museum and research complex located in Washington D.C. It opened in 1910 and today houses about 3,75,000 mineral specimens from around the world. It is one of the largest collections of its kind with great value to the scientific community. The purpose of the collection is to support scientific research.



Summary :

- Minerals are identified and described by observation and analytical methods. There are about 5,389 mineral species recognised.
- Minerals derive their names after their founders, localities where they were found first or from their chemical composition and physical properties.
- Depending upon chemical composition and atomic structure, minerals exhibit different physical properties like colour, streak, lustre, form, cleavage, fracture, hardness and specific gravity. These are readily used to identify and differentiate minerals.
- Based on chemical composition, minerals are classified as oxides, sulphates, halides, carbonates and silicates. Silicates are most abundant and are classified according to their structures as nesosilicates, sorosilicates, cyclosilicates, phyllosilicates and tectosilicates.
- Rocks are aggregates of minerals which give us clues to the rock forming processes.

Q. 11. Long answer questions :

- 1) Discuss in detail the criteria for classifying a substance as a mineral.
- 2) Give the interdependence of cleavage, hardness and lustre citing suitable examples.
- 3) Enumerate and describe the forms dependant on the degree of crystallization. Give suitable examples.
- 4) Describe with examples, forms independent of the degree of crystallization.

Activity no. 1 : Observe the photographs, identify and distinguish between the following pairs :



Activity no. 2 : Crossword

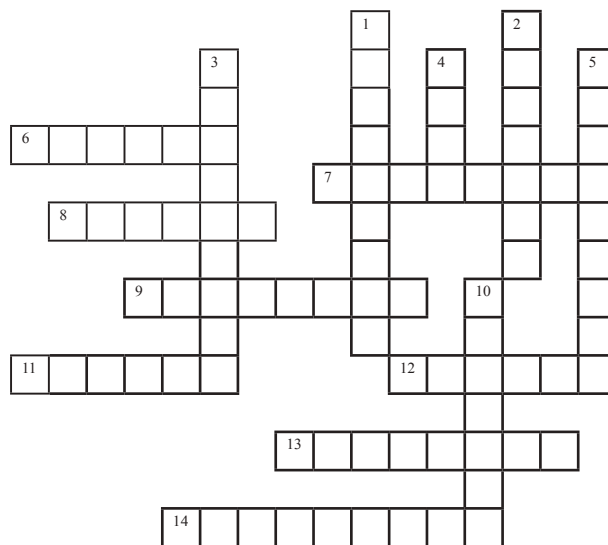
Name the common minerals, using the given information.

DOWN :

- 1) a naturally magnetic mineral
- 2) a bright blue copper carbonate
- 3) metamorphic quartz sandstone
- 4) a silicate mineral known for its perfect cleavage resulting in thin sheets
- 5) strontium sulphate its name means celestial
- 10) a common carbonate mineral often a replacement mineral in fossils

ACROSS :

- 6) the main source of lead
- 7) an iron ore, sometimes used in jewellery
- 8) sometimes called fool's gold
- 9) sulphide mineral, long metallic slender bladed crystals
- 11) a very heavy sulphate mineral
- 12) rock salt
- 13) calcium fluoride
- 14) a bright green copper carbonate



Activity no. 3 : Observe the back cover of your text book and answer the following questions :

- 1) Name the 'Father of Mineralogy'.
- 2) Classify the stamps into their mineral classes.
- 3) List the minerals found within the state of Maharashtra.
- 4) Ice is considered as a mineral - Justify.
- 5) Identify and name the native elements depicted in the stamps.
- 6) Sort the minerals containing (OH) group in their chemical composition.
- 7) Categorise the minerals having 'Water of Crystallisation' in their chemical composition.
- 8) Find and name all the minerals with 'Ca' in their chemical composition.
- 9) List all the minerals which are oxides.
- 10) Enumerate all the carbonate minerals.

Activity no. 4 : Collect the information on minerals museums in the state of Maharashtra/India.



Highlights :

- An account of physiography and stratigraphy of Maharashtra
- Geographic and geological distribution of economic mineral deposits of Maharashtra.

Introduction :

After gaining sufficient knowledge on the fundamental aspects of geology, it is necessary to understand the geology of our own state. Geology of Maharashtra occupies special status globally due to Deccan volcanism and the Lonar meteorite crater. The state is also gifted with economic minerals and hydrocarbons. Maharashtra is the third largest

state in area after Rajasthan and Madhya Pradesh. Maharashtra covers a large part of west-central India between North latitudes $15^{\circ}45'$ and $22^{\circ}00'$, and East longitudes $73^{\circ}00'$ and $80^{\circ}59'$. The total area of Maharashtra is 3,07,713 sq. km. exposing a wide range of geological features and mineral resources. Rock formations from Archaean to Holocene age are exposed in Maharashtra.

6.1 Physiographic subdivisions of Maharashtra :

Maharashtra can be divided into three physiographic regions (Fig. 6.1), namely-

- 1) Maharashtra Plateau
- 2) Western Ghats (Sahyadri)
- 3) Coastal Region (Konkan)

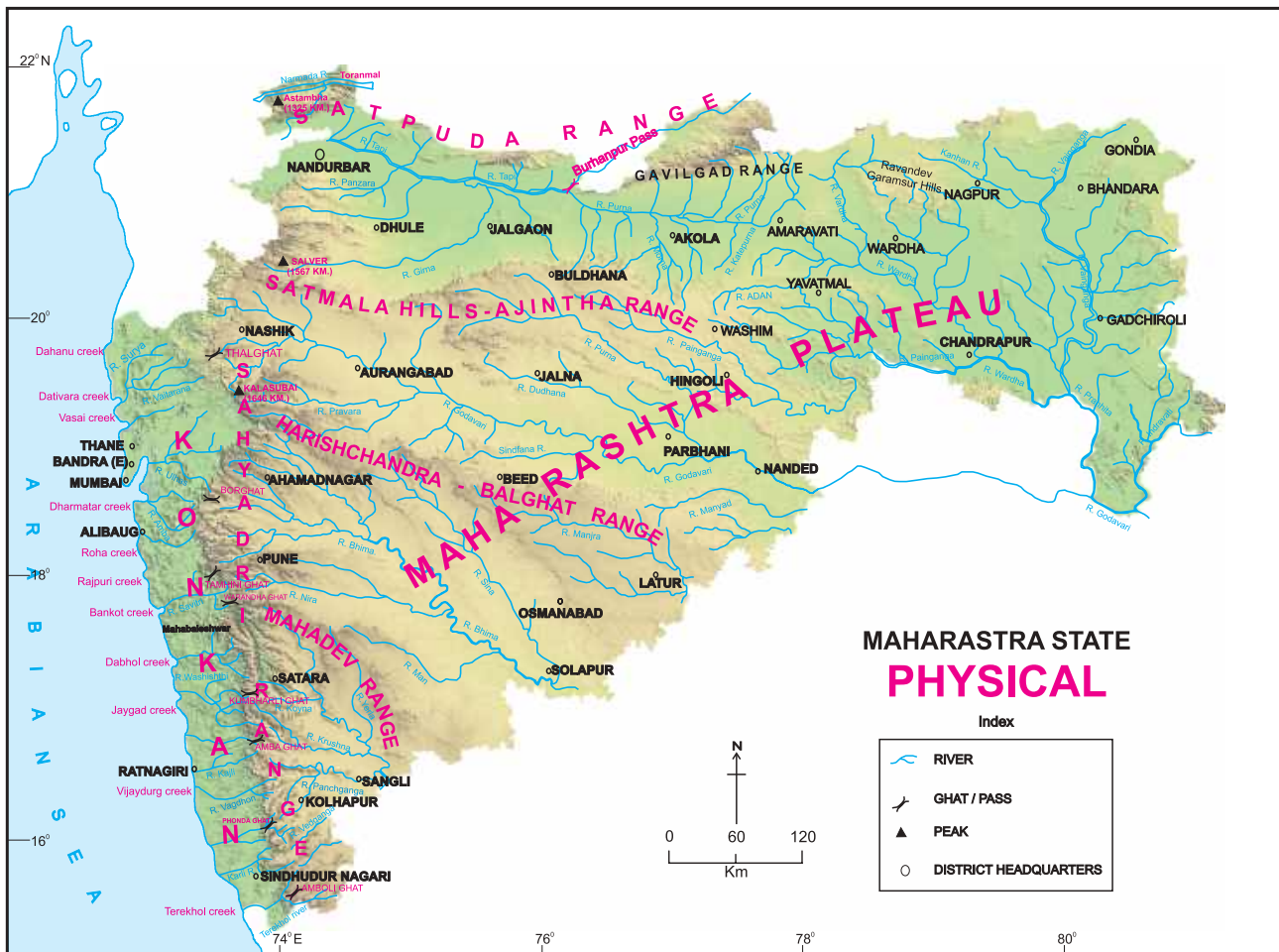


Fig. 6.1 : Map showing the physiographic features of the state of Maharashtra.

1) Maharashtra plateau : Maharashtra Plateau is a major part of Deccan plateau of Peninsular region of India. It is formed by basaltic lava erupted through fissures. These basaltic lava flows are horizontal to sub-horizontal and are found to spread practically all over the state. The landscape is more or less uniform with flat-topped hill ranges. They occupy about 81% area of the state and mainly covers Vidarbha, Khandesh, Marathwada and some Western Maharashtra divisions. The region is traversed by east-west and north-west to south-east trending river valleys, and hill ranges with more or less similar trends. The altitude of Maharashtra Plateau varies from 400 to 600 msl with gentle eastward gradient. It has been exposed to denudation for several millions years.

The Satmala, Ajantha, Harishchandra-Balaghat range and Mahadev hills stretch across the Maharashtra plateau. Their altitude ranges between 600m and 900m above msl. Melghat range which is a branch of Satpura hill trends east-west, almost parallel to northern border of the State. Satmala-Ajantha hills trend in east-west direction and separate Tapi basin from Godavari basin. Harishchandra-Balaghat hills trending north-west-south-east separate Godavari basin and Bhima basin. Mahadev hills run north-west-south-east, dividing Bhima and Krishna basins.

2) Western Ghats : The Western Ghats are also known as Sahyadris. They are situated to the west of Maharashtra Plateau and extend north-south. They are parallel to the west coast of India. The average elevation is between 1000 and 1300 msl and decreases from south to north. Many hills are places of pilgrimage, hill stations and forts.

The topography is formed by differential weathering and erosion of rock masses. The rock masses are composed of series of

near horizontal lava flows of basalt. The thickness of the lava flows varies from less than a meter to more than 35m. Due to differential weathering and erosion, harder lava flows form steep scarps whereas the softer lava flows form gentler slopes, resulting in a step-like topography. Because of this step-like or terraced appearance the hills are called 'Traps'. These hills are generally flat-topped or table-topped. The Sahyadris have a gentler slope towards the Deccan plateau and steep slope towards west. Major rivers in Maharashtra flow in an easterly to southeasterly direction with their sources in Sahyadri, whereas some of their tributaries originate in the east-west trending ranges. The major easterly flowing rivers are Godavari and Krishna which originate at Trimbakeshwar and Mahabaleshwar respectively, in the Western Ghats. Godavari basin is the largest basin of the Indian Peninsular region next to Ganga basin. Wardha, Penganga, Wainganga, Manjra, Pranhita, Indravati and Shabari are the main tributaries of Godavari. Koyna and Bhima are the major tributaries of Krishna. The major westerly flowing rivers are Narmada and Tapi. The Narmada river originates at Amarkantak in Madhya Pradesh and flows along the northern boundary of Maharashtra. Whereas the Tapi river originates near Multai in Betul district of Madhya Pradesh and flows nearly parallel to the Narmada river before meeting the Arabian sea.

Do you know?

The highest peak in the Sahyadris is Kalsubai at 1646m (5400ft) which is located in the Harishchandragad-Kalsubai sanctuary. Kalsubai is also known as the Everest of Maharashtra.

3) Coastal region : A narrow coastal plain between Western Ghat and Arabian Sea is known as Konkan coast. In Maharashtra

state, the coastal lowland is called 'Konkan' and the vast plateau region east of the Ghats is known as 'Desh'. The Konkan coastal belt is about 720 km long and its width varies from about 50 - 105 km. The formation of coastal lowland mainly depends on the length of rivers, volume of water (perennial or seasonal) flowing through rivers, the alluvium brought by rivers, extent of catchment areas and slope of land.

The Konkan belt consists of number of east-west trending ridges. There are some small and narrow plateaus. The average height of these plateaus is about 60 to 100m above msl. These plateaus are covered by laterite in South Konkan. Several short and swift rivers originate in Sahyadri and flow westwards finally to enter the Arabian Sea. Due to limited length and heavy rainfall, the velocity of water flow is high. Many of the coastal streams carry

out severe erosion in their source region and have caused diversion of some of the plateau rivers, towards west. This has resulted in development of waterfalls all along the western slope of Sahyadri. The major rivers are Ulhas, Vashishthi, Shastri, Vaitarna, Savitri, Patalganga, Kundalika and Terekhol.

6.2 Stratigraphy :

Various rock formations are exposed in the state of Maharashtra along with their varied lithology. The distribution of different rock formations is shown in (Fig. 6.2) and their stratigraphic succession is given in Table 6.1.

- i) **Paleo-archean (3600 - 3200 Ma) :** The oldest rocks exposed in the state are biotite gneisses belonging to Amgaon gneissic complex, Tirodi gneisses and Peninsular gneisses, that formed the basement.

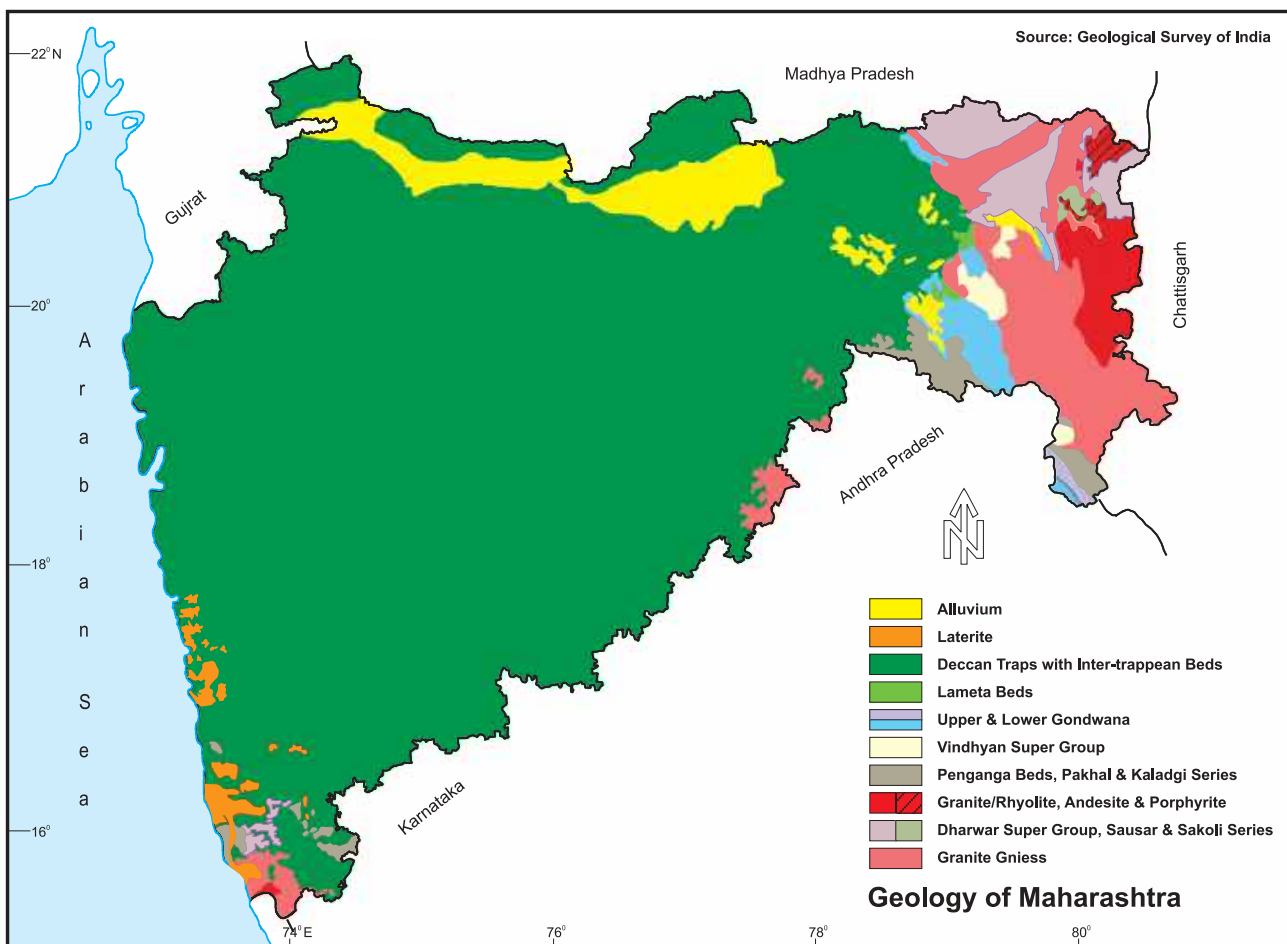


Fig. 6.2 : The generalized Geological map of Maharashtra

ii) Neo-archean (2800 - 2500 Ma) : The basement gneisses are overlain by volcanics and metasediments belonging to Amgaon, Sakoli and Dongargarh groups. They are represented by different types of schists, gneisses, quartzites, marbles, amphibolites, hornblende schists with intercalated bands of iron and manganese ores. These rocks were involved in several orogenic cycles and hence are intensely deformed and mixed up. They are intruded by Dongargarh granite, a batholith in Gondia and Gadchiroli districts.

iii) Paleo-to-Meso proterozoic (2500 - 1000 Ma) : After the intrusive activity there was a long period of non-deposition during which these rocks were subjected to weathering. This resulted in the development of Eparchean unconformity. Bhandara, Nagpur and Gondia regions are represented by Khairagarh and Sausar groups. The Sausar group hosts one of the richest manganese ore deposits.

iv) Neo-proterozoic (1000 - 541 Ma) : Denuded topography formed during Eparchean unconformity became the sites of deposition for the Neo-proterozoic sedimentation. They are represented by sediments of Kaladgi Group in the Ratnagiri district and sediments of Pakhal Group, Vindhyan Supergroup and Penganga Group in the Gadchiroli, Chandrapur and Yavatmal districts.

During the period between Cambrian and Upper Carboniferous there is no lithological record in Maharashtra. During this period India was part of Gondwanaland and was covered by thick sheet of ice.

v) Upper Carboniferous (323 - 299 Ma) : During Carboniferous period glacial to fluvio-glacial sediments were deposited in the linear rift basins, that are exposed in Nagpur, Chandrapur and Yavatmal districts represented by 'Boulder Bed' or tillites at the base (Talchir Formation).

vi) Permian (299 - 252 Ma) : Talchirs are followed upward by deposition of fluvial sediments during warmer period now represented by coal-bearing sandstones and shales of Barakar and Kamthi formations.

vii) Triassic (252 - 201 Ma) : The sedimentary sequence viz., Mangli, Pachmari and Maleri formations are represented by sandstone and shale.

viii) Jurassic (201 - 145 Ma) : This period is represented by the Kota and Chikiala formations comprising of limestone.

Do you know?

The strata ranging in age from Upper Carboniferous to Jurassic is called as Gondwana Supergroup.

ix) Upper Cretaceous (100 - 66 Ma) : After the cessation of sedimentation in the Gondwana basin around 100 Ma, there was a marine incursion marked by deposition of Bagh beds in Dhule district and contemporaneous fluvial and lacustrine Lameta beds were deposited in parts of Nagpur, Chandrapur, Amravati and Yavatmal districts.

x) Upper Cretaceous to Lower Paleocene (~ 69 - 61 Ma): This time is represented by voluminous outpouring of lava through extensive fissures resulting in the formation of Deccan basalt, which is one of the well-known flood basalt provinces of the world. The Deccan traps cover an area > 500,000 sq. km. The eruption was mostly non explosive and the lava spread over extensive areas in the form of nearly horizontal flows of varying thickness (Fig. 6.3). Number of such flows are found piled one above the other and some individual flows can be traced for several hundred kilometers. The flows occupy almost the entire area of Maharashtra except some parts of Nagpur, Bhandara, Gondia, Chandrapur, Gadchiroli and Yavatmal

districts in Vidarbha region and Ratnagiri and Sindhudurg districts in coastal Konkan. Two types of lava flows are recognized. They are the pahoehoe or ropy lava and the 'aa' or blocky lava. Pahoehoe lava flows predominate in Dhule, Aurangabad, Pune and Nasik districts, whereas in rest of Maharashtra, 'aa' lava flows are dominantly exposed. In Maharashtra between some flows intertrappean beds of lacustrine origin are found. **Red bole beds** commonly occur in between two lava flows. The flows are intruded by number of dolerite dykes trending nearly N-S, parallel to the west coast. East-west trending dykes are commonly exposed in northern part of Maharashtra around Nandurbar, Dhule and Jalgaon districts.



Fig. 6.3 : Stratigraphic exposures of lava flows in Malshej Ghat area of Maharashtra

Do you know?

The term 'Deccan trap' was first coined by W. W. Sykes in 1833 after the Swedish word 'trapp / trappa' meaning stair to describe the step-like or terrace-like topography peculiar to Deccan trap terrain. The name Deccan is an anglicised form of the Prakrit word 'Dakkhini', itself derived from the Sankrit word 'Dakhina' meaning south.

Deccan basalts have uniform mineralogical and chemical characters. It is essentially composed of plagioclase and augite with

accessory amount of biotite and hornblende. Olivine may or may not be present. It shows compact, vesicular, amygdaloidal, pipe amygdaloidal, pillow and columnar structure. The cavities in vesicular basalts are filled with secondary minerals like heulandite stilbite, apophyllite, quartz, amethyst, chalcedony, agate, jasper, opal, calcite etc. Ropy lava structure, columnar joints are commonly observed in basalts (Fig.6.4 and 6.5).



Fig. 6.4 : Ropy Lava



Fig. 6.5 : Columnar joints, Tumzai Hills, Panhala

Do you know?

The rock cut caves of Ajanta, Ellora and Elephanta are the UNESCO world heritage sites.

The Ajanta caves located in Aurangabad are 30 rock cut Buddhist cave monuments in Deccan Basalt.

The Kailash Temple or cave 16 as it is known at Ellora is a huge monolithic temple dedicated to Lord Shiva.

ash occurrence are unique to the Quaternary sequence of Maharashtra.

6.3 Distribution of major economic minerals of Maharashtra :

Though most of the mineral deposits of Maharashtra state are concentrated in eastern Maharashtra, the major mineral deposits of bauxite occurs in western Maharashtra. The minerals which are being economically exploited are manganese ore, iron ore, coal, bauxite, limestone, sillimanite, kyanite and clays. Deposits of copper, gold, tungsten, barite, magnetite, ilmenite and chromite are also worth mentioning. Other minerals like andalusite, antimony, asbestos, lead and zinc, platinum, silver, diamond, corundum and radioactive minerals are occurrences that have not been proved to be economically viable so far.

6.3.1 Manganese ore :

Manganese ore deposits of Maharashtra are the part of world famous manganese ore producing Nagpur-Bhandara-Balaghat belt of India. Manganese is mainly used in the manufacture of steel. It is also used in the manufacture of ferro-alloys, dry batteries, manganese chemicals and paints.

6.3.2 Iron ore :

Iron ore deposits of Maharashtra occur in Chandrapur, Gadchiroli, Gondia and Sindhudurg districts. They are mostly concentrated in the Vidarbha region. The iron ore minerals are hematite and magnetite.

6.3.3 Coal :

Coal is a compact, stratified rock formed by accumulation and preservation of plant matter. It is combustible and considered a fossil fuel. Plants when buried in marshy land get decayed and carbonization process gives rise to coal. These coal bearing areas are present only in the Vidarbha region. Coal is being mined from

Table 6.2 : Lithostratigraphy of Deccan traps
(M1 - M4 are marker horizons)

Super Group	Group	Sub Group	Formation
D E C C A N T R A P S	S A H Y A D R I	Mahabaleshwar	Mahabaleshwar
			M4
		Diveghat	Purandargad
			Diveghat
			Elephanta
		Lonavala	Karla
			Indrayani
		Kalsubai	M3
			Upper Ratangad
			M2
			Lower Ratangad
			M1
			Salher

xi) Paleocene-Oligocene (66 - 23 Ma) : This period is mainly represented by lateritization in western Maharashtra.

xii) Miocene-Pliocene (23 - 2.58 Ma) : This period represents episodes of sedimentation and lateritization.

xiii) Pleistocene-Recent (<2.58 Ma) : This period is represented by Quaternary alluvium, soils and laterites. Important archaeological sites and the Toba volcanic

several underground and opencast mines in Nagpur, Chandrapur and Yavatmal districts. Major portion of coal produced is used for power generation, while part of coal produced is utilized in cement plants, textile mills, paper mills, brick and lime kilns and railways. Coal deposits in Vidarbha have the potential of meeting the requirements of increasing thermal power generation and other industries in western Maharashtra.

Lignite (low rank brown coal) also occurs in association with Tertiary sediments in Ratnagiri and Sindhudurg districts of Maharashtra. It is brown in colour and contains abundant non-combustible matter.

6.3.4 Bauxite :

Bauxite is a mixture of various minerals of aluminium hydroxide. It is a porous clay-like rock sometimes having pisolitic structure, and different colours depending upon its composition. The colours generally vary between earthy white to brownish yellow. Bauxite occurs as blanket deposits, lenses and pockets and intercalated beds within sedimentary strata or as detrital deposits. It also occurs as a layer below laterite cap developed on basalts of Sahyadri. Bauxite is principally used as a source of aluminium, but it is also used in refractory, chemical, abrasive, iron and steel, cement and other industries.

Bauxite deposits in Maharashtra are reported from Kolhapur, Raigad, Satara, Sangli, Sindhudurg, Ratnagiri and Thane districts.

6.3.5 Limestone and dolomite :

Limestone is a fine grained sedimentary rock predominantly composed of calcium carbonate. In Maharashtra limestone occurs in rock formations of different ages, which includes, Kaladgi, Pakhal, Vindhyan, Penganga, Gondwana, Lameta, Bagh beds and intertrappean beds. However, economically

important limestones occur in Neoproterozoic Vindhyan Supergroup and Penganga Group of Yavatmal and Chandrapur districts. In these districts, considerable reserves of cement and flux grade limestones are available. A good quality limestone is mainly used in cement industry. It is also used in paper, sugar, rubber, metallurgical and other industries.

Extensive deposits of dolomite associated with metamorphic rocks of Sausar Group occur in Nagpur district. Dolomite is also associated with limestones of Penganga Group in Chandrapur and Yavatmal districts.

6.3.6 Kyanite and sillimanite :

These are aluminium silicate minerals commonly found in metamorphic rocks like gneisses and schists. In Maharashtra kyanite and sillimanite are used as refractory material in refractory industries because of their high degree of melting point and alumina content. They are also used for the manufacture of porcelain used in spark plugs. Kyanite and sillimanite deposits in Maharashtra are present in Bhandara and Gadchiroli districts.

6.3.7 Fluorite :

Occurrence of fluorite is reported mostly from Dongargaon area of Chandrapur district. Fluorite along with barite occur as cavity filling deposits. It is being economically exploited by Maharashtra State Mining Corporation. It is used in metallurgy, glass and ceramic industries.

6.3.8 Barite :

Significant occurrences of barite have been reported from Mul area of Chandrapur district. Some minor occurrences are also noticed in Gadchiroli, Yavatmal and Sindhudurg districts. Barite occurs as cavity filling mineral in quartz veins in granite gneiss. It is used in drilling mud, filler in paper and rubber industries.

6.3.9 Chromite :

Chromite is the ore of chromium. In Maharashtra, chromite occurs in Bhandara, Chandrapur, Nagpur and Sindhudurg districts. Chromite occurs as pockets, veins, lenses and as discontinuous bodies. It is used in steel, refractory and metallurgical industries.

6.3.10 Copper :

Copper mineralisation occurs in shear zones traversing the Sakoli Group of rocks from Nagpur, Bhandara, Chandrapur and Gadchiroli districts. The deposits in Chandrapur, Gadchiroli and Nagpur districts are of economic importance.

6.3.11 Tungsten :

Tungsten in the form of wolframite and sheelite mineralization occurs in Nagpur district. Minor occurrences are reported from other parts of Nagpur and Bhandara districts.

6.3.12 Silica sand :

Silica sand is available from the sandstones and decomposed quartzites belonging to Kaladgi Supergroup occur in south Konkan. The sand deposits mainly occur in Sindhudurg, Ratnagiri and Kolhapur districts at number of places. These reserves are used in foundry, chemical and glass industries.

6.3.13 Secondary minerals :

Zeolites and associated secondary minerals including varieties of silica, occur as cavity filling minerals in Deccan Trap. They include heulandite, stilbite, apophyllite, natrolite, mesolite, cavanzite, chabazite, prehnite etc. Silica minerals include rock crystal, amethyst, agate, opal etc.

6.3.14 Oil and natural gas :

The off-shore area along the west coast of Maharashtra has assumed importance because

of the occurrence of oil and natural gas. The off-shore area includes the giant Bombay High Field which is surrounded by several small fields. The Bombay High, which is the largest known oil field in India, is located about 160 km west-north-west of Mumbai in the Arabian Sea. In Maharashtra, potential areas for Coal Bed Methane (CBM) have been identified in Wardha Valley coal field areas.

Summary :

- Physiographically Maharashtra state is divided into three regions :
a) Maharashtra Plateau, b) Western Ghats and c) Coastal region.
- Rock formations from Archean to Holocene age are exposed in Maharashtra.
- Archean rocks belonging to Amgaon gneissic complex, Tirodi gneisses, peninsular gneisses form the basement. Basement gneisses are overlain by volcanics and meta-sediments. These formations have been later intruded by dolerite dykes at many places. After the Eparchaeon unconformity, rocks of Neoproterozoic period were deposited. These are overlain by basaltic rocks which erupted during Cretaceous period and cover about 81% of the state.
- Coal deposits are found in Vidharbha region. Commercial deposits of non-metallic minerals like limestone and dolomite, kyanite, sillimanite, fluorite, barite, etc. are found in southern and eastern regions of Maharashtra. Off shore areas of Maharashtra state have the commercial deposits of oil and natural gas. These are being exploited from 'Bombay High'.

Table 6.1 : Stratigraphic sequence in Maharashtra state.

Stratigraphic Sequence	Age in Million Yrs.	Representative Rock Formations	Geographic Distribution
xiii) Holocene-Pleistocene (Quaternary)	< 2.58	Alluvium, laterite, sand, soils, volcanic ashe	Younger and older alluvium in Nagpur, Bhandara, Gondia, Chandrapur, Wardha, Yavatmal, Akola, Amravati, Jalgaon, Dhule and Nandurbar districts. Laterite in Kolhapur, Satara, Sangli, Raigad, Ratnagiri, Sindhudurg and Thane districts. River terraces of Wainganga, Wardha and Penganga rivers; raised beaches along west coast
xii) Miocene-Pliocene	2.58 - 23.0	Sediments, lignite, shales and laterites	Ratnagiri and Sindhudurg districts
xi) Paleocene-Miocene	< 66	Laterites, saprolites, ferricretes, bauxites, lateritic soils	Laterites occur in majority of coastal stretches. Few occurrences are their in central eastern Maharashtra
x) Palaeocene-Upper Cretaceous	~ 69 - 61	Deccan Trap basalt flows with intertrappean beds	Basalt flows cover most of the state from west of Nagpur and Chandrapur, up to the Arabian sea coast excepting in the eastern parts of Nagpur, Bhandara, Gondia, Chandrapur, Gadchiroli and Ratnagiri districts. Intertrappeans occur in Nagpur, Yavatmal, Chandrapur districts and Mumbai.
ix) Upper Cretaceous	100 - 66	Infratrappes (Lameta and Bagh beds)	Lameta Formation in Nagpur, Chandrapur and Yavatmal districts. Bagh Beds in Dhule district.
viii) Jurassic	201 - 145	Limestone of Chikiala and Kota formations	Gadchiroli district
vii) Triassic	252 - 201	Clays and sandstone of Maleri and Pachmari formations	Sironcha Tahsil of Gadchiroli district and Achalpur Tahsil of Amravati district
vi) Permian	299 - 252	Sandstone and shale (Mangli Formation) Sandstone and shale (Kamthi Formation) Sandstone, shale and coal (Barakar Formation)	Nagpur, Chandrapur, Yavatmal and Gadchiroli districts
v) Upper Carboniferous	323 - 299	Talchir Formation	Nagpur, Chandrapur and Yavatmal districts
iv) Neo-proterozoic	1000 - 541	Limestone, shale and sandstone (Vindhyan Supergroup) Limestone and shale (Penganga Formation)	Nagpur, Chandrapur and Yavatmal districts
		Limestone and shale (Pakhal Group)	Chandrapur district
		Conglomerate, sandstone and shale (Kaladgi Group)	Ratnagiri and Sindhudurg districts
iii) Palaeo - Meso-proterozoic	2500 - 1000	Khairagarh Group, Sausar Group	Nagpur, Bhandara and Gondia districts
ii) Neo-archean	2800 - 2500	Dongargarh Granite, Nandgaon Group, Sakoli Group, Bailadila Group, Amgaon Group	Nagpur, Bhandara, Gondia, Chandrapur and Gadchiroli districts
i) Paleo-archean	3600 - 3200	Tirodi and Amgaon Gneissic Complex, Peninsular Gneissic Complex	Nagpur, Bhandara, Chandrapur, Gadchiroli, Ratnagiri and Sindhudurg districts.

EXERCISE

Q. 1. Fill in the blanks :

- 1) Bagh beds occur in districts.
 - a) Dhule b) Ratnagiri
 - c) Satara d) Hingoli
- 2) Lameta beds occur in parts of districts.
 - a) Yavatmal b) Kolhapur
 - c) Solapur d) Raigad

Q. 2. True or false :

- 1) During the period between Cambrian and Upper Carboniferous there is no lithological record in Maharashtra.
- 2) A narrow coastal plain between Western Ghat and Arabian Sea is known as Konkan coast.
- 3) Sahyadris are also known as Western Ghat.

Q. 3. Choose the correct alternative :

- 1) The pahoe-hoe lava flows predominate in
 - a) Dhule - Aurangabad- Pune- Nasik
 - b) Osmanabad- Sangli- Satara- Latur
 - c) Nagpur - Buldhana- Amravati- Akola
 - d) Parbhani - Beed- Nanded- Akola

Q. 4. Match the following :

River	Origin
a) Narmada	i) Betul
b) Godavari	ii) Amarkantak
c) Krishna	iii) Trimbakeshwar
d) Tapi	iv) Mahabaleshwar
1) a-ii,	b-iii, c- iv, d-i
2) a-iii,	b-ii, c-i, d- iv
3) a- iv,	b-i, c-ii, d-iii
4) a-i,	b- iv, c-iii, d-ii

Q.5. Arrange the following Geological formations of Maharashtra in the correct stratigraphic sequence (oldest to youngest) :

- a) Lameta beds - Talchir formation - Kaladgi group
- b) Talchir formation- Kaladgi group- Lameta beds
- c) Kaladgi group - Talchir formation- Lameta beds
- d) Lameta beds -Kaladgi group - Talchir formation

Q. 6. Give geological reasons :

- a) Iron mining is economically suitable in Maharashtra.
- b) Bauxite mining is feasible in Maharashtra.
- c) Secondary minerals like zeolite occur in Deccan traps.

- d) Thermal power plants in Maharashtra have their coal source in Vidarbha region.

Q. 7. Write Short notes on :

- 1) Building stones of Maharashtra.
- 2) Energy resources in Maharashtra.
- 3) Secondary minerals found in Deccan Traps.
- 4) Uses of kyanite and sillimanite.
- 5) Occurrence of following economic deposit.
 - a) Manganese b) Iron ore c) Coal
 - d) Bauxite e) Chromite f) Copper

Q. 8. Answer in detail :

- 1) Describe lithostratigraphy of the Deccan traps.
- 2) Give the genesis of Laterite and Bauxite.
- 3) Explain why thickness of Basalt is variable in the Deccan Volcanic Province .
- 4) Give an account of major economic minerals of Maharashtra.
- 5) Describe general succession of stratigraphy of Maharashtra.

Q. 9. Read the following passage and answer the questions :

The Singhbhum Shear Zone hosts mineral deposits of great economic value. The sulphide deposits were formed in this shear zone before the third-phase deformation. The original copper mineralization is attributed to hydrothermal activity. The passage of hydrothermal fluids was controlled by faults of the rift zone. Along the shear zone, the deposits of chalcopyrite, pyrite and pyrrhotite are syngenetic with respect to volcano - sedimentary phase, predating the main tectonothermal event. The total reserve in the Singhbhum Shear Zone is of the order of 217 million tonnes with 1.36 % Cu. In the Gangpur Basin, the lead and copper deposits of the Sargipali belt are estimated to be 206 million tonnes with 6.73 % Pb and 0.33 % Cu.

Ref : K S Valdiya, *The making of India* 2016.

- 1) What kind of deposit is characteristic of the Singhbhum Shear Zone?
- 2) How is the copper mineralization associated in the shear zone?
- 3) Calculate the total estimated Cu and Pb content in the Singhbhum Shear Zone.
- 4) If the Sargipali deposits are dated 1660 Ma, find the period referring to the geological time scale.

